FactoryTalk Optix v1.8

Technical Overview

September 2026

Rockwell Automation

expanding **human possibility**®

PUBLIC

Agenda

01 Core Capabilities

02 Expanded Architecture

03 ThinManager 14.2 & FactoryTalk Optix 1.8

04 Digital Design Experience

05 Data Connectivity

06 Extended Runtime Licensing

07 Firmware 7.0.0.x for Optix-based devices

08 2026 FactoryTalk Optix webinar series

09 Online Demos

10 PCDC Release Notifications

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FactoryTalk® Optix™ Customer-Driven Capability Themes

The customer needs shaping our roadmap — and why each one matters



General / Customer Features

Comprehensive hardware and software offerings to meet your application demands

Continued prioritization of your requests

Why?
Because your success is the foundation of ours. Thank you!



Regulatory Compliance

- Machine Regulations
- Cyber Resiliency Act (CRA)
- OPC UA Certification
- Cybersecurity Certifications

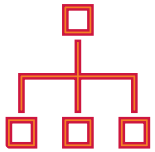
Why?
So that your systems meet global safety and cybersecurity standards with uninterrupted market access.



Gen AI / Co-Pilot Assistance

Helping you benefit from the power of AI with smarter design and operations assistance

Why?
So that you can create better applications faster and keep your machines running longer more efficiently.



Expanded Architectures

Helping you make sense of all the data in your system

Managing and securing systems with many assets

Supporting larger systems

Why?
As technology evolves, we want to be the OT company that your IT department wants to work with.

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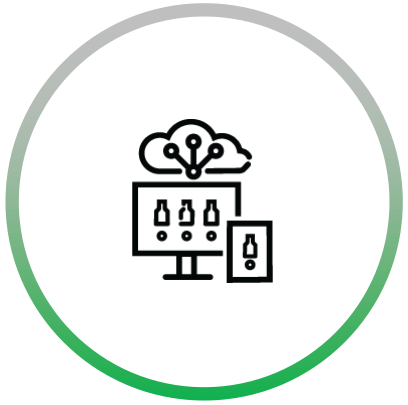
FactoryTalk® Optix™ Development Priorities

The four pillars we build against — how the themes become engineering priorities



Core Capabilities

- Core product capabilities that create a **strong** and **sustainable** platform for core automation **visualization** and **edge applications**.



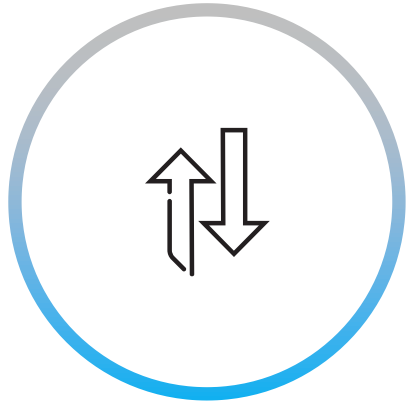
Expanded Architectures

- Support for **integrated hardware**, larger distributed **architectures**, process **systems**, remote control, and **cloud**-based operations.



Digital Design Experience

- **FactoryTalk Hub** integrated workflows for a cohesive customer experience when building a system using **SaaS-based design tools**.



Data Connectivity

- Robust **edge connectivity** and application platform to enable data and **analytics** with core **operations** data.

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FactoryTalk® Optix™ v1.8 — Focus Areas Delivered

What ships in v1.8 against each development focus area



Core Capabilities

- Retentivity Backup & Restore
- Download and Upload files from the Web
- Report DataGrid Word Wrap
- RA EtherNet/IP API for design-time tag browsing and import
- Recipe Schema APIs for design-time configuration



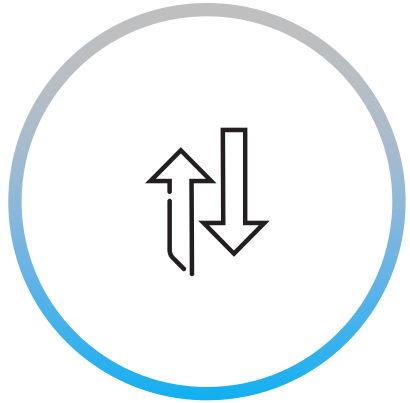
Expanded Architectures

- High Availability Runtime cluster
- Workstation Client
- ThinManager user SSO into Optix application
- New Runtime Licensing - Base + Extended



Digital Design Experience

- AI Design Assistance
- FactoryTalk Design Studio integration
  - Offline tag import



Data Connectivity

- MQTT Enhancements
  - Complex datatypes support
  - Event log publishing
- RA Device Driver for direct connectivity to PowerFlex drives

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Core Capabilities

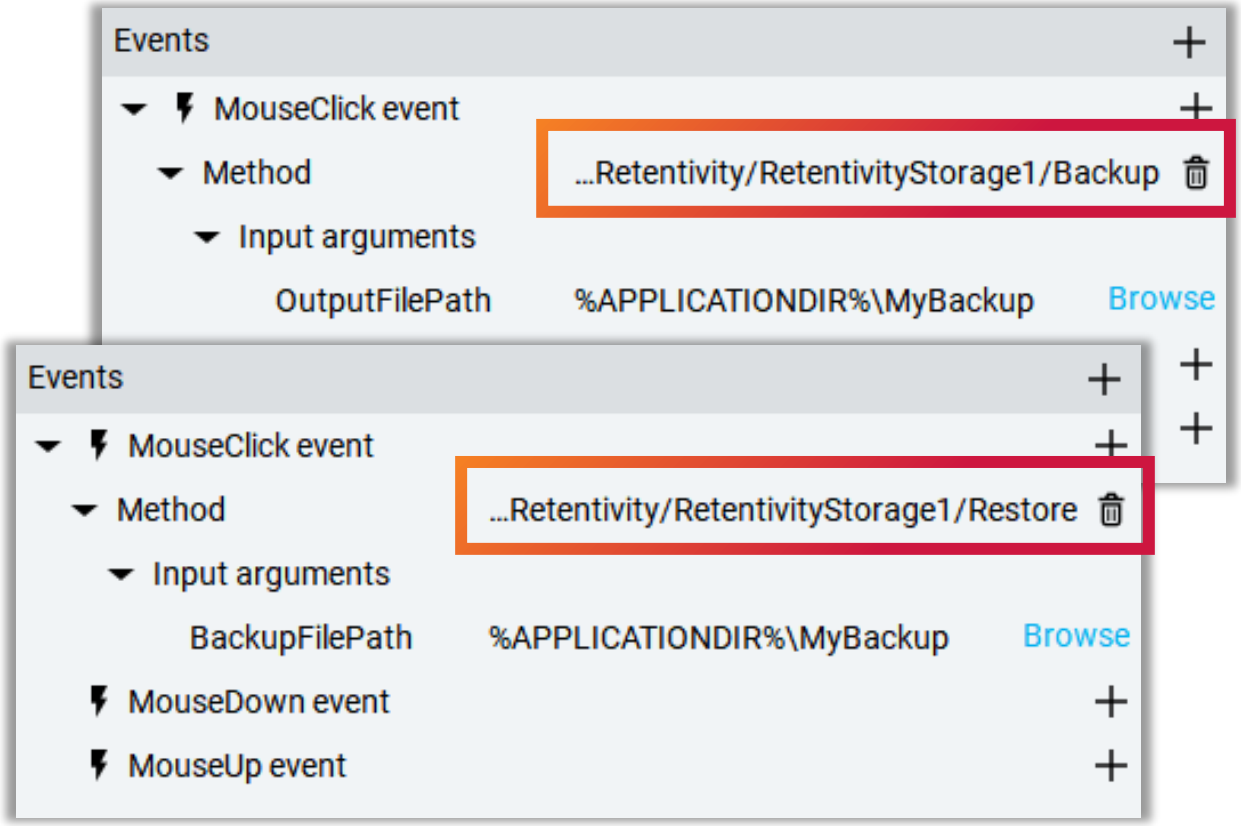
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Retentivity Backup and Restore

Core Capabilities

The **Retentivity Storage** now provides **Backup** and **Restore** methods:

- **Save** the current retentive data state to a **backup file**
- **Restore** retentive data from an **existing backup**



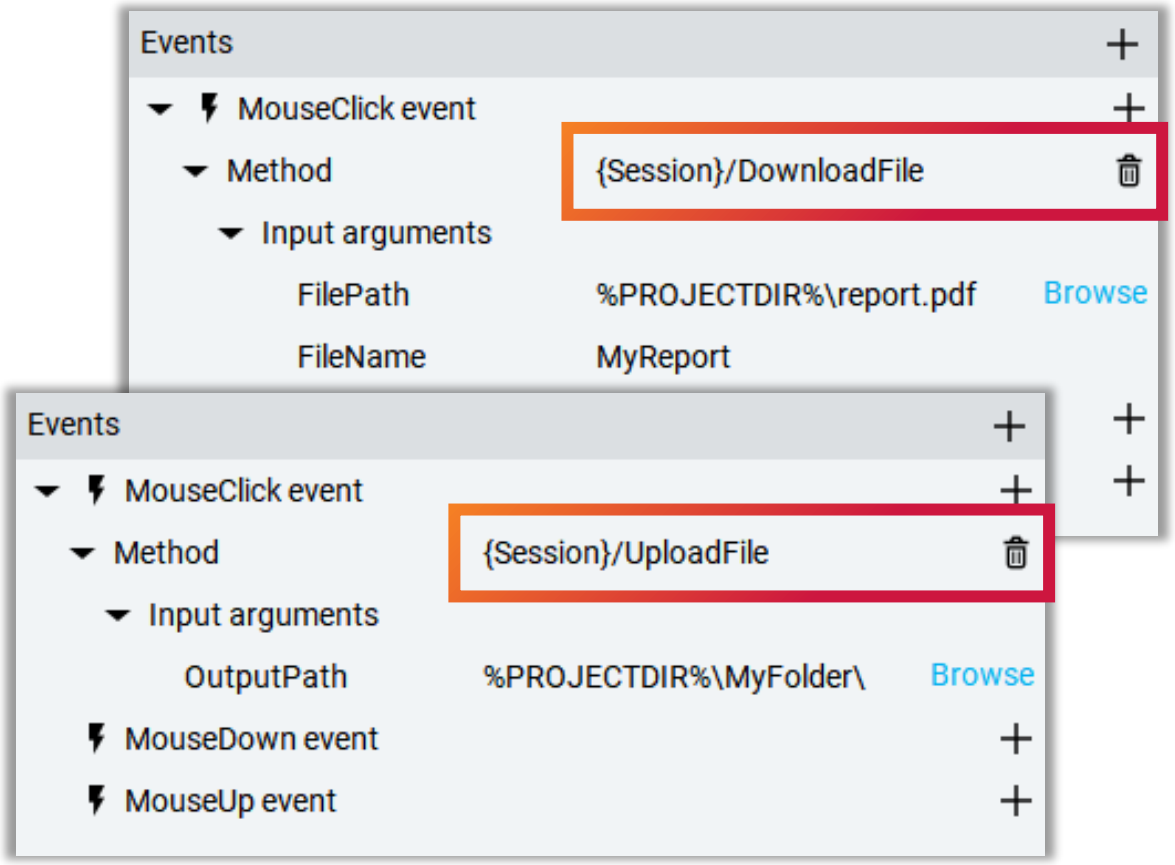
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Download and Upload from Web

Core Capabilities

Added **Download and Upload methods** to the UI Session:

- **Download files** from the Runtime to the **client device**
- **Upload files** from the client device to the **Runtime**



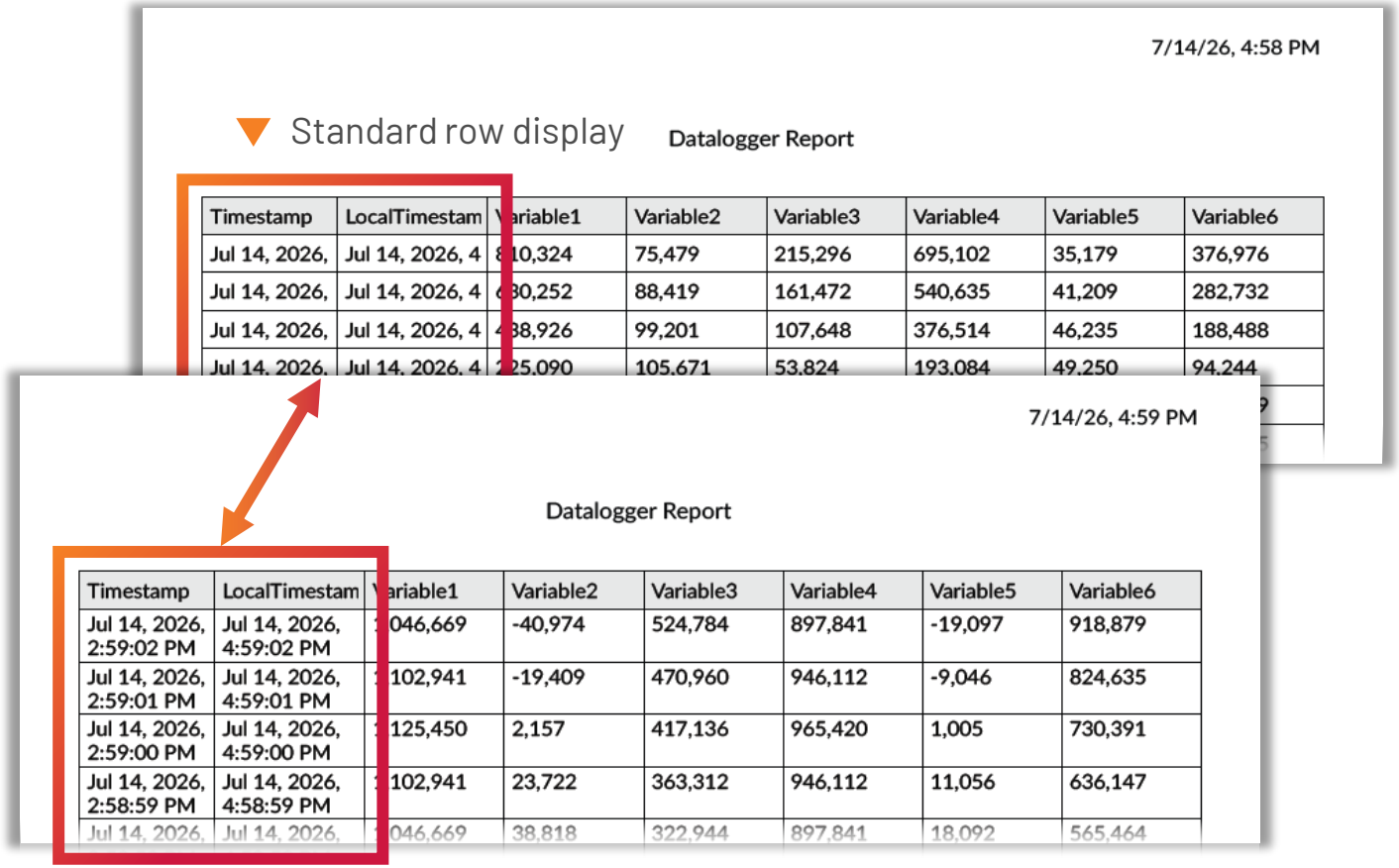
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Readable Reports for Long-form Data

Core Capabilities

New **word wrap** control in the Report DataGrid:

- **Present** long-form entries such as timestamps, batch records, audit logs, and operator notes **in full**
- Give report authors **direct control** over how cell content flows across **multiple lines**



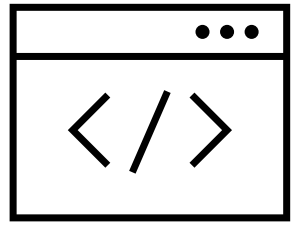
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New C# APIs

Core Capabilities

1- Added an **RA EtherNet/IP API** for design-time tag **browsing and import**:

- **Browse** controller tags programmatically
- **Import** tags at design time using NetLogic



2- Added **APIs for** design-time **Recipe Schema configuration**:

- **Create** and **modify** Recipe Schemas using NetLogic
- **Automate** recipe setup and maintenance

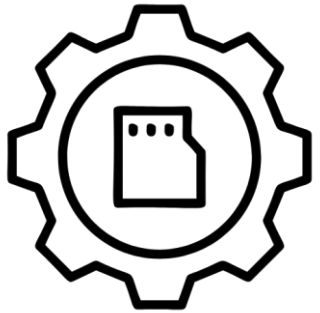
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Memory Optimization

Core Capabilities

1- Describe COW / COW benefits:

- Bullet 1
- Bullet 2



2- Describe memory allocate/deallocate benefits:

- Bullet 1
- Bullet 2

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Expanded Architectures

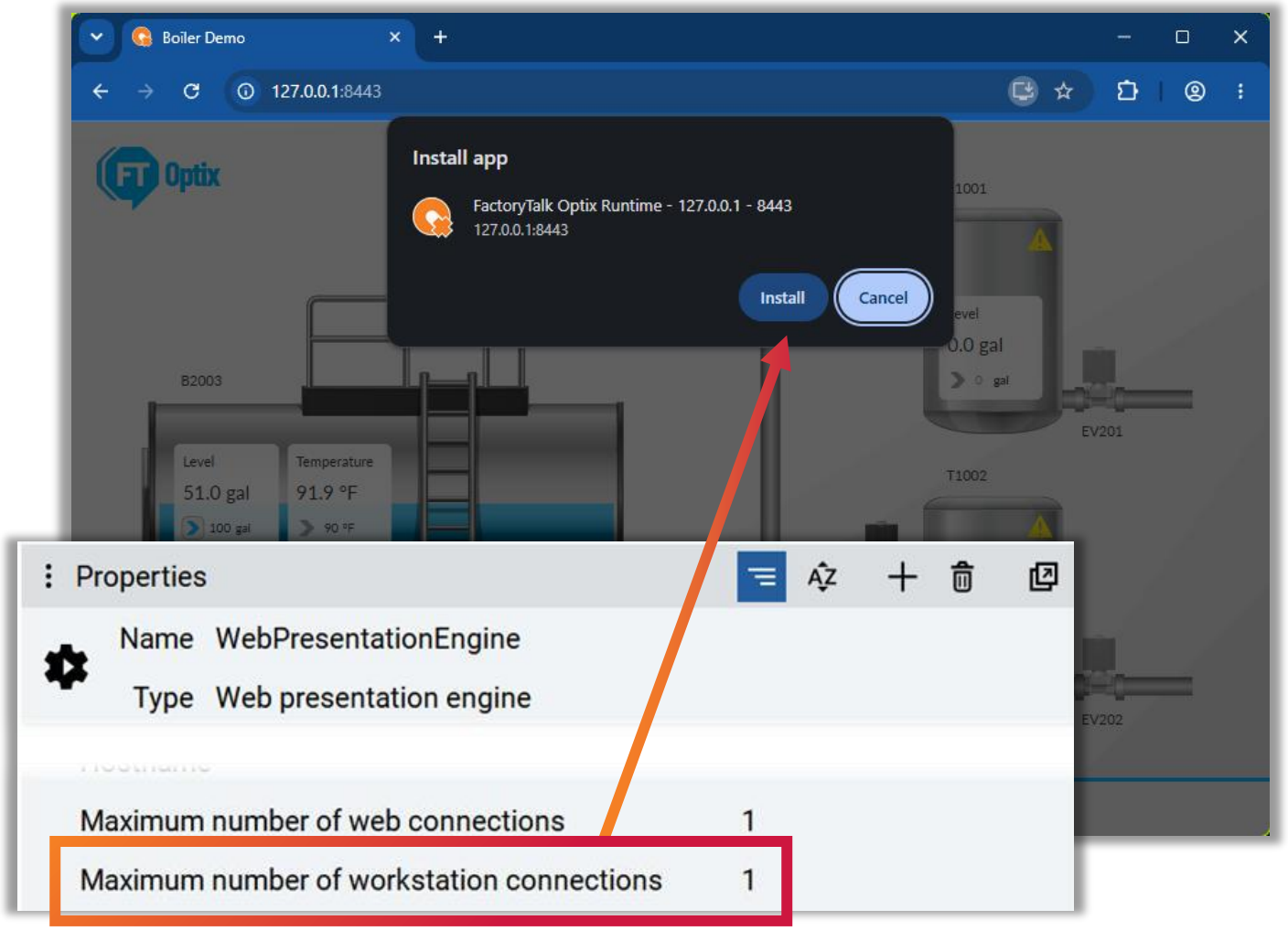
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Workstation Clients

Expanded Architectures

Run FactoryTalk Optix applications as **workstation clients** installed directly from the browser:

- **Configure** the **maximum number** of workstation client connections
- **Install** the **application** on a workstation device
- **Launch it** directly from the desktop



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FactoryTalk Optix High Availability (HA) Overview

What is High Availability

- High Availability is the ability of a FactoryTalk Optix application to remain operational and accessible for a long period.
- It is achieved by deploying an application to a three-node high availability runtime cluster, ensuring that if the primary node fails, a standby node immediately takes over to prevent downtime and keep the application running.

Core concepts of High Availability

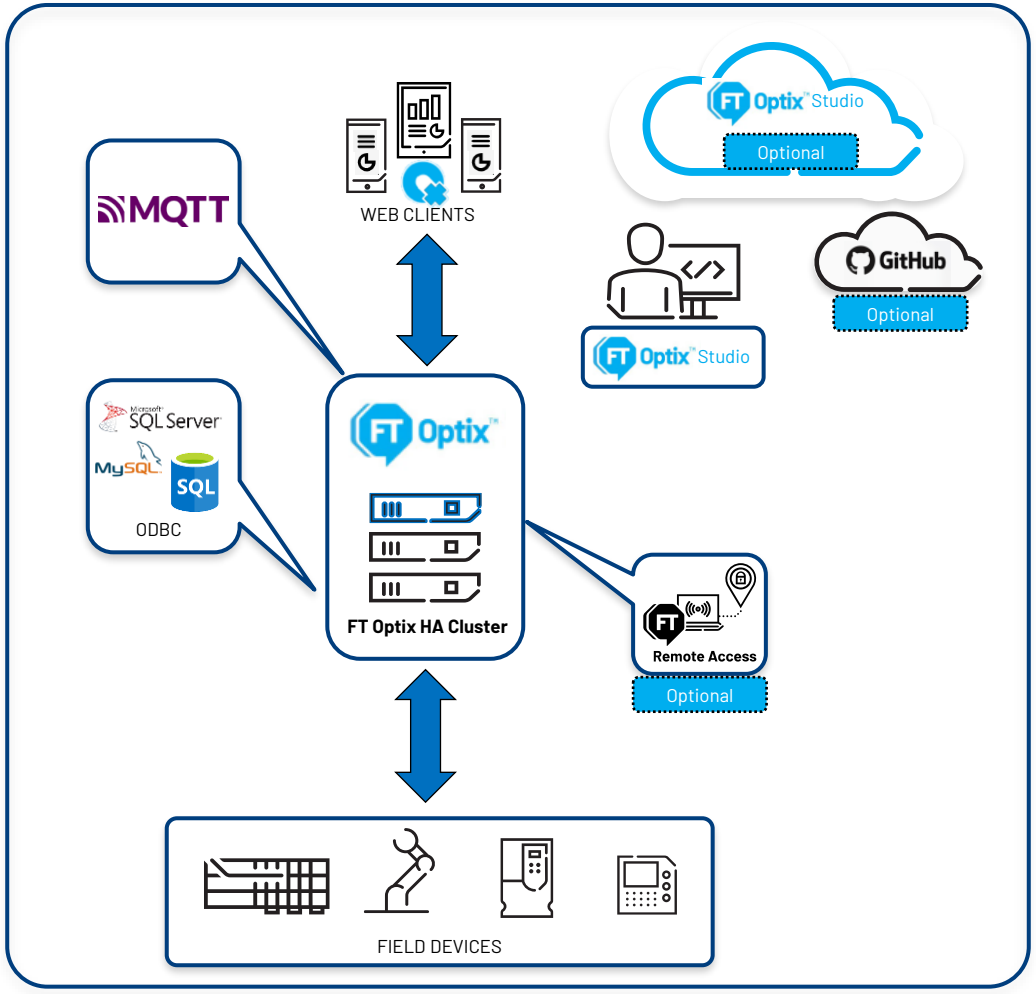
- Three-node High Availability Runtime Cluster: A group of three interconnected servers (nodes) that run the FactoryTalk Optix application with synchronized data.
- Automated Failover: The process of automatically switching to a standby node if the primary node fails or goes offline.
- Fault Tolerance: The ability for the application to continue functioning when a node or software component fails.

Why High Availability Matters

- Ensures the FactoryTalk Optix application remains continuously operational and accessible.
- Guarantees stakeholders have reliable, real-time access to critical insights.
- Helps meet system uptime percentages over a designated period - referred to as “nines” (99.99%).

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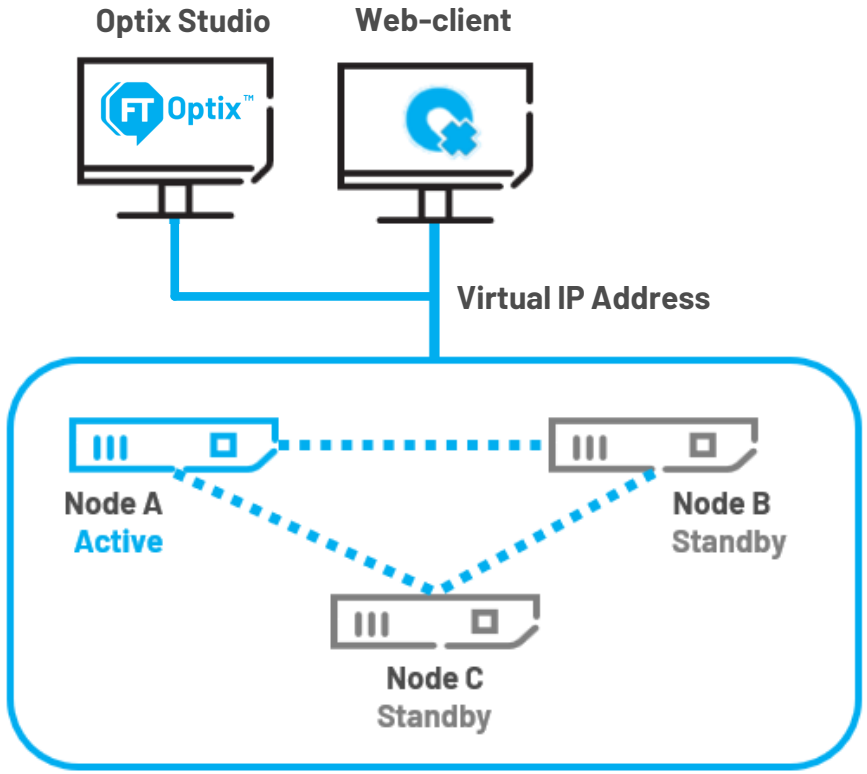
FactoryTalk Optix High Availability Runtime Cluster



Three-node Highly Available Runtime Cluster

FactoryTalk Optix runtime data synchronization and failover to standby server

- Runtime application is deployed to a three-node high availability cluster
- One active node runs the application, with two standby nodes ready for automatic failover
- Data is replicated across three separate nodes to ensure availability
- System can tolerate a node failure without losing access to data or services
- Each node holds a full copy of critical data, allowing seamless failover from active to standby node
- Automatic resynchronization occurs when a failed node comes back online
- Clients connect via a virtual IP and automatically reconnect to the active node during failover



Three-node high availability runtime cluster

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Key Benefits of a 3-node Highly Available Cluster

Recommended for maximum reliability and uptime

Maximize uptime by design

Continued reliable operation even after one node is unavailable.

No split-brain risk

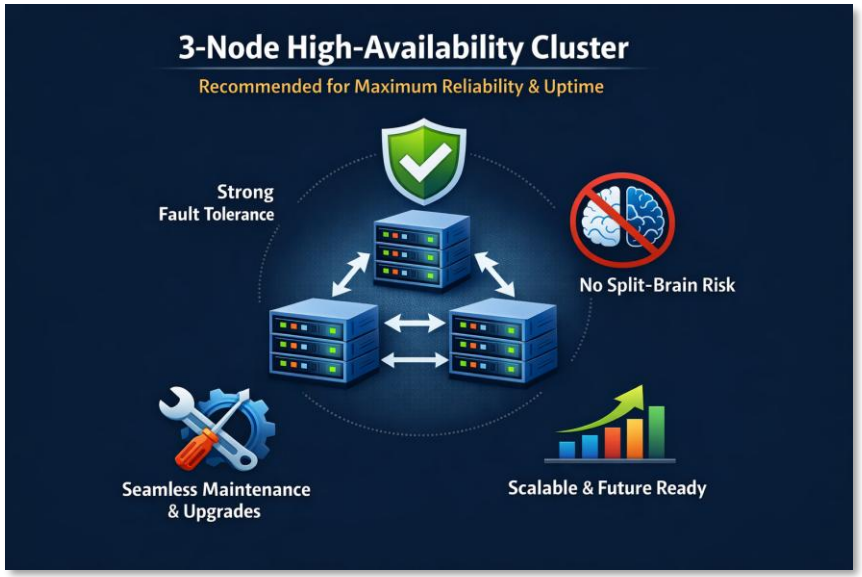
A clear primary runtime is always maintained across the cluster.

Industry standard for modern architecture

Best practice and recommended solution providing superior reliability and maximizes uptime

Ready to scale

Supports load balancing as Optix grows to larger architectures (planned).

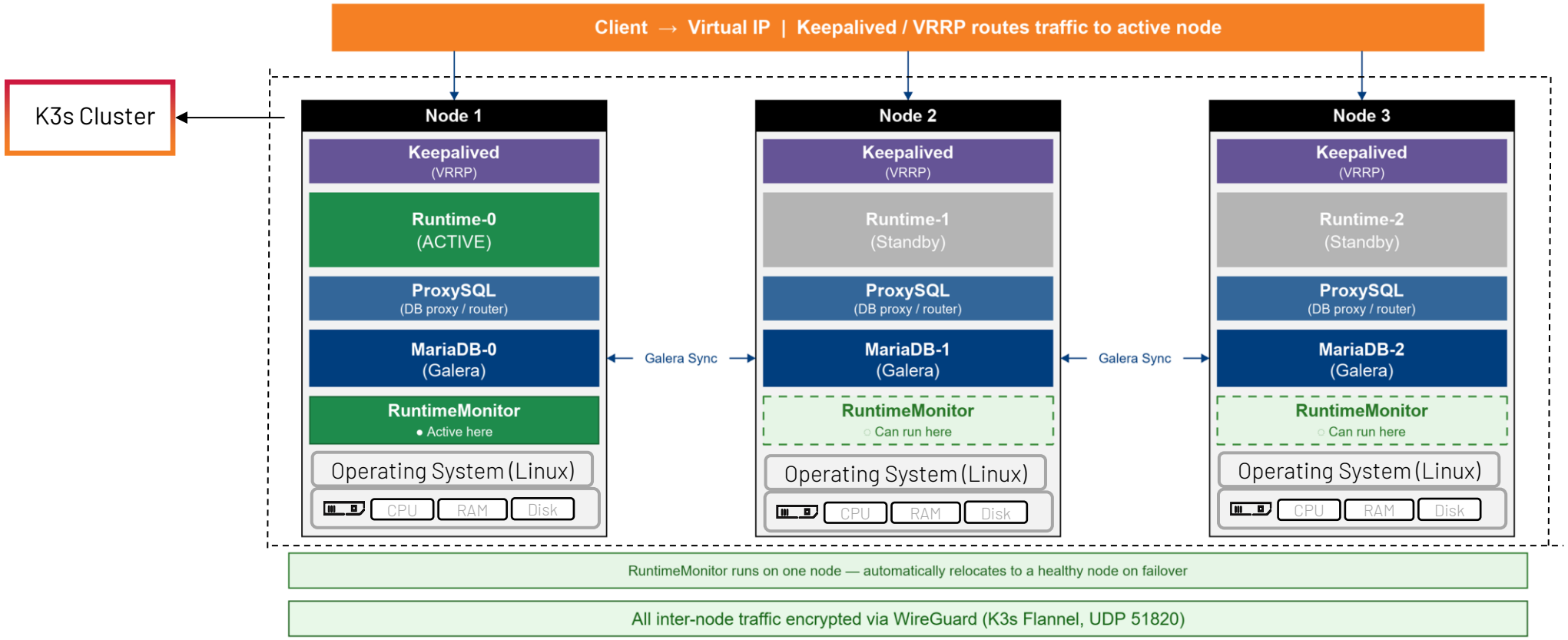


"FactoryTalk Optix High Availability is built to keep systems running smoothly through failures, maintenance, and growth."

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High Availability Architecture

Always available. Always synchronized. Always ready.



Additional capabilities may be added to each release as priorities are refined.

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Kubernetes Overview

Kubernetes is an open-source platform that automates deploying, scaling, and managing containerized applications

CORE CAPABILITIES

Load balancing
Spreads network traffic evenly so no single part of the app gets overloaded

Self-healing
Automatically restarts dead or failing containers and replaces unhealthy ones

Auto-scaling
Grows or shrinks the number of running app copies based on how much work or traffic there is

K3s A fully certified, lightweight, easy-to-install Kubernetes distribution for resource-constrained environments. The Optix high availability solution is designed to support any Kubernetes compliant cluster with version 1.34.6 or later.

KEY CONCEPTS

Cluster A group of physical or virtual machines working together to run applications

Node An individual computer (worker machine) inside the cluster

Pod The smallest group of containers running your app together on a node

Control Plane The central brain that watches the system and decides where pods should run

Industry standard A well-proven, *de facto* platform for container orchestration

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FactoryTalk Optix High Availability Manager

A high availability cluster install and management tool

1 Pre-requisites

- Three physical PCs to form the cluster
- Install Linux OS (native or virtual)
- Network / IP address configuration
- Obtain virtual IP address

2 Install cluster package – complete delivery bundle of container images and charts

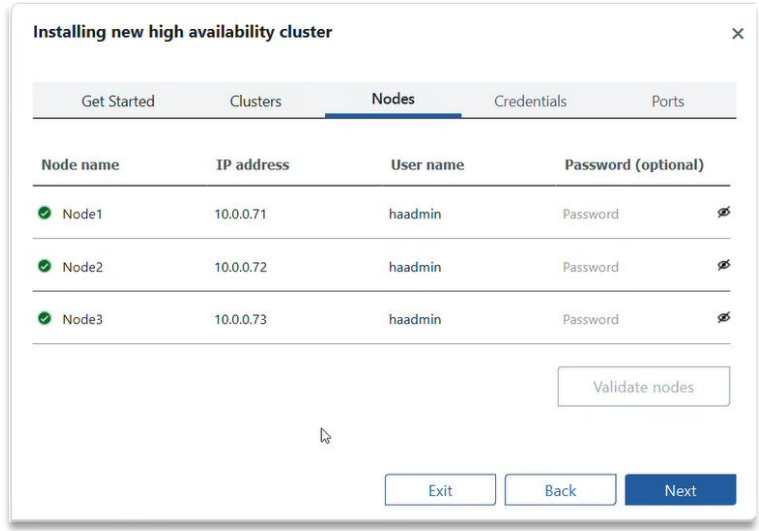
- Keepalived for virtual IP management
- K3s Lightweight Kubernetes for container orchestration
- FactoryTalk Optix Installable Runtime
- RuntimeMonitor and UpdateServer for health, status and coordinated updates
- MariaDB Galera cluster for synchronized, fault-tolerant data replication

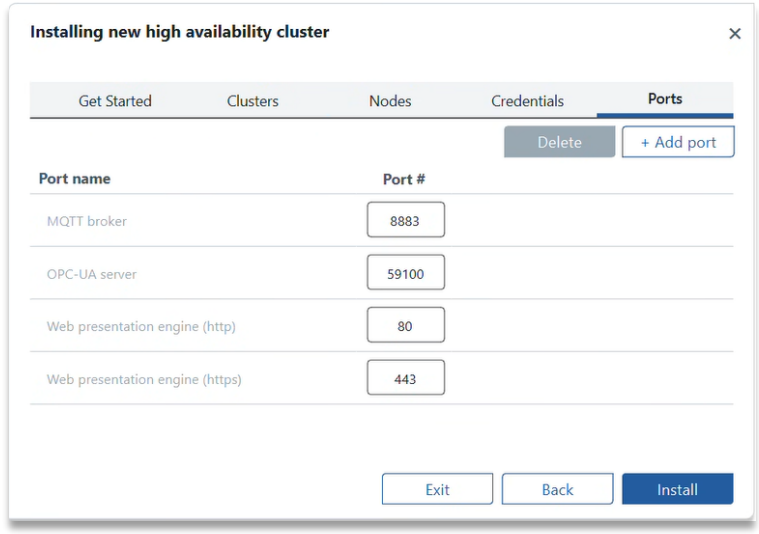
3 Port configuration

- MQTT, OPC-UA and Web presentation engine ports

4 Manage entitlements

- Base and extended licenses





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FactoryTalk Optix v1.8 High Availability

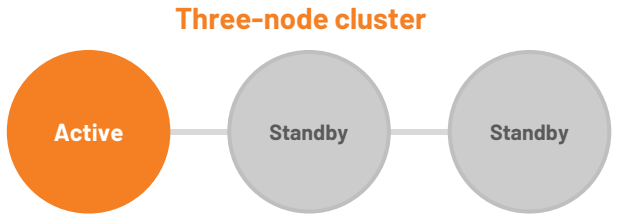
Without High Availability

- ✕ Server down means the application and all clients are down
- ✕ Recovery takes days
- ✕ Days of lost visibility and data
- ✕ Replicating server is costly (2 x Optix Runtime entitlement)

With High Availability

- ✓ Application and clients switch automatically to the next active node
- ✓ Recovery in seconds or minutes
- ✓ Minimal visibility and data loss
- ✓ Cost effective (1 x Optix Runtime entitlement)

How it will be implemented



- A new deployment option for Optix Runtime.
- Enable high availability in the Optix project.
- Deploy to a three-node cluster with the Optix HA Manager.
- One active node, two standby nodes for automatic failover.
- Application replicated across all three nodes.
- Tolerates a node failure with minimal loss of service.
- Clients reconnect automatically during failover.

Takeaways

- Built to keep systems running through failures, maintenance, and growth.
- Proven off-the-shelf technology (Kubernetes, containerized deployment) with simple configuration.
- Existing single-server runtimes upgrade easily to an HA deployment using extended feature tokens.
- Reach out to your local Rockwell Automation or distributor contact with any other questions.

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ThinManager 14.2 + FactoryTalk Optix 1.8

Containerized applications managed by ThinManager

Docker container technology to run and visualize applications

OT-friendly ThinManager tools make containers easy to use and visualize

Remove maintenance of and investment in server-based resources all together by running ThinManager on an IPC and applications in containers

Web-browser container

- Firefox, Chrome, Citrix StoreFront
- Access web applications without requiring RDS-hosted browsers
- Patch browsers quickly and easily through ThinManager



FactoryTalk Optix runtime container

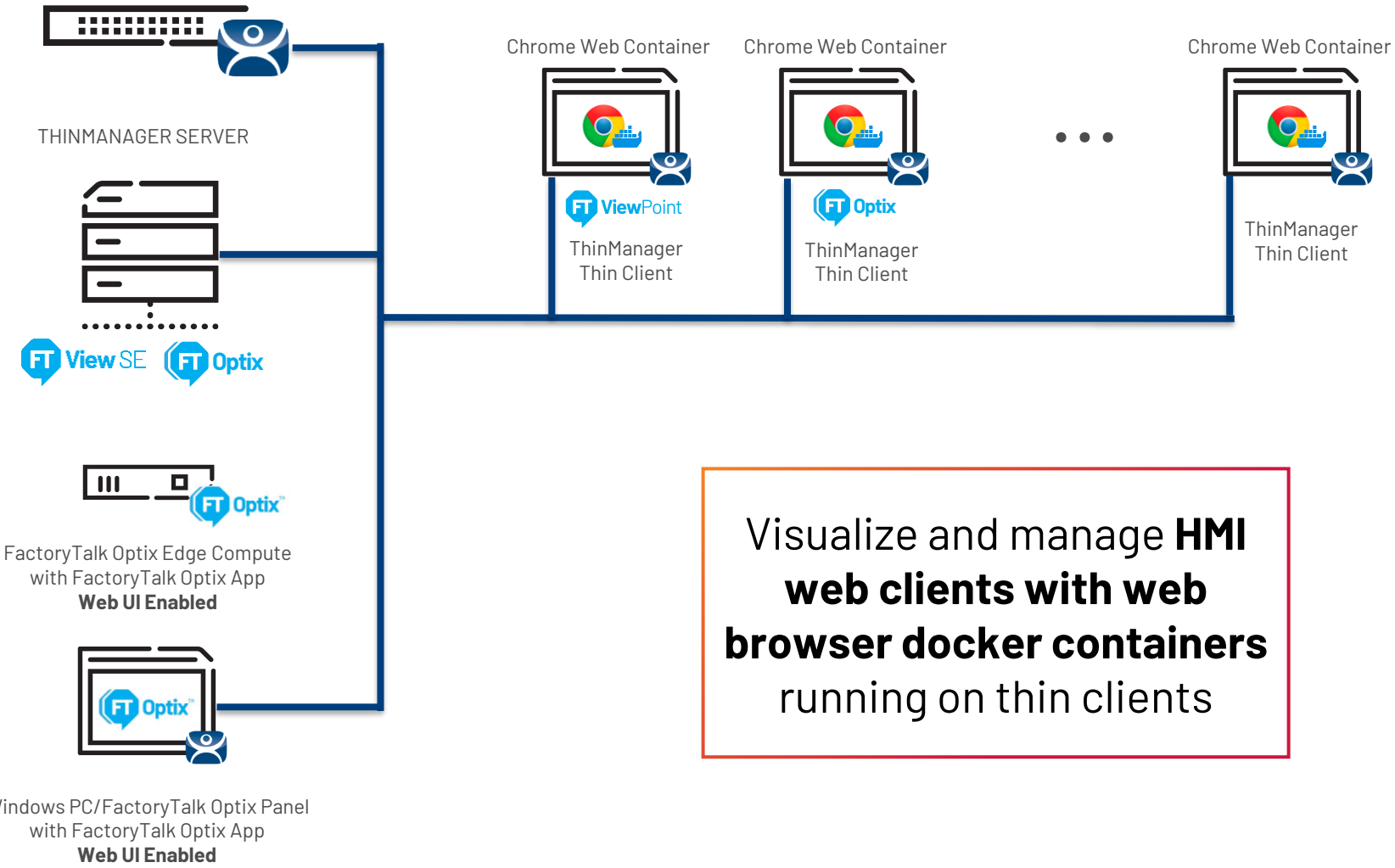
- Export application from Optix Studio and utilize Native Presentation Engine to run container on thin client
- Deploy, manage and scale FT Optix applications through ThinManager



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Managed web clients

Utilize ThinManager Chrome or Firefox docker container to connect to web-based software

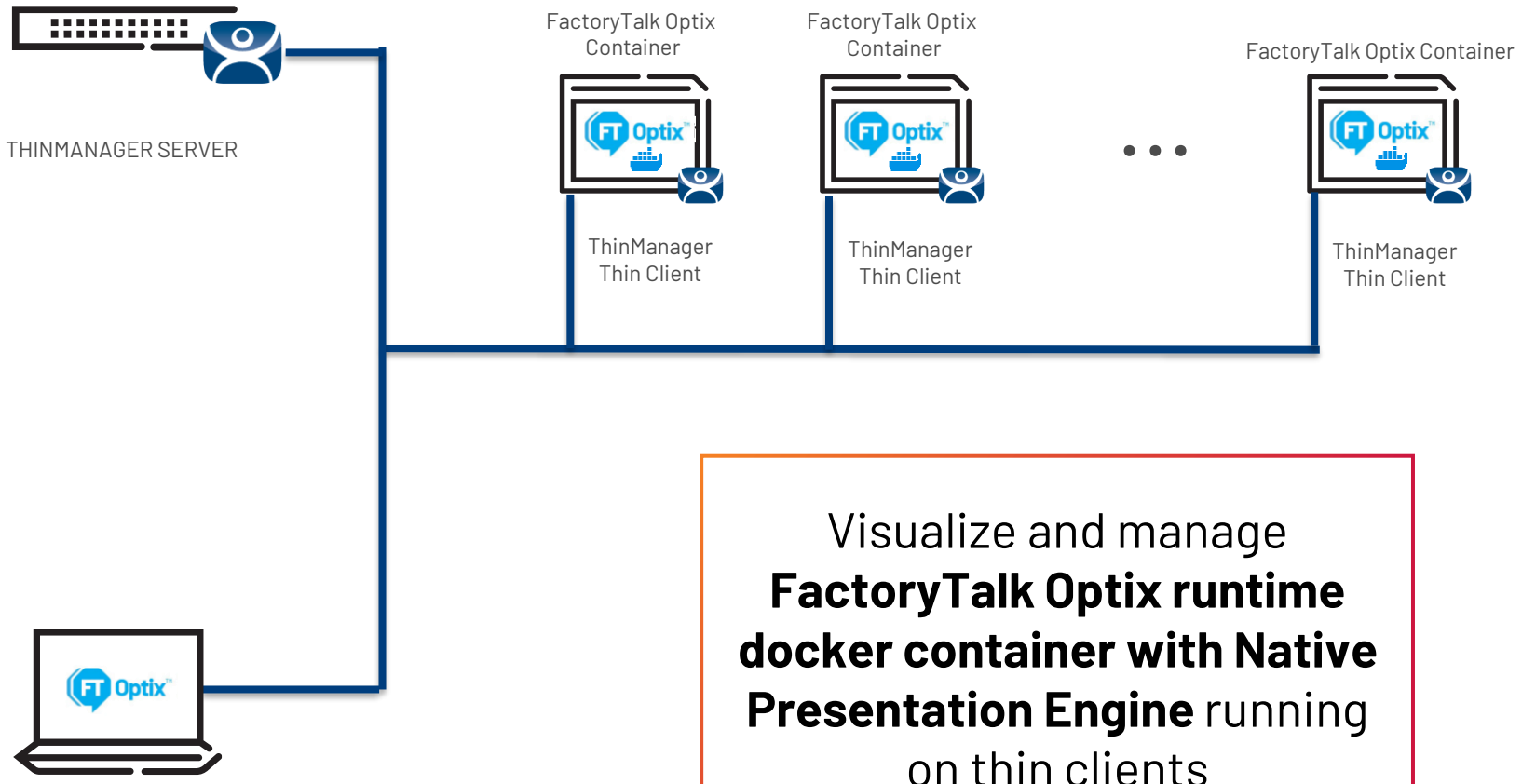


Visualize and manage **HMI web clients with web browser docker containers** running on thin clients

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Managed runtime containers

Utilize FactoryTalk Optix Native Presentation Engine in ThinManager docker container



Visualize and manage **FactoryTalk Optix runtime docker container with Native Presentation Engine** running on thin clients

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Value of Managing FactoryTalk Optix with ThinManager

- **SSO functionality requires Optix 1.8 and installation** of the FactoryTalk Optix ThinManager extension into Optix Studio (available on ThinManager.com following Optix 1.8 release)
- **Choose hardware** with FactoryTalk Hub for Optix Runtime and Device Sizing Tool
- **Properly license** each terminal with an Optix Container Display Client that will require unique Optix entitlement key
- **Connect thin clients to the internet** as Optix 1.6+ and ThinManager 14.2 only supports online entitlements for Docker Container deployments
- **Utilize the FactoryTalk Optix Native Presentation Engine** from the ThinManager-hosted Optix docker container

Key considerations

- Finish development and **deploy final version** of Optix HMI application
- **Store and manage multiple versions** of your Optix application in ThinManager
- **Scale** your solution and **deploy** applications to visualize on **multiple ThinManager terminals**
- **Speed up boot times** with ThinManager + ASEM local storage option BootAssist™ to cache container image and Optix application on thin client

Key benefits

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Managing FactoryTalk® Optix™ with ThinManager

For customers who want secure, centrally managed FactoryTalk® Optix™ applications

Increased Security

Reduce cybersecurity footprint with **zero clients** running docker containers utilizing local device memory

Double-insulated web browser **isolated** within a Docker Container hosted on **closed** Linux firmware

Reduce IT Management

Simplify firmware **updates** and patch **management** of browsers and applications

Reduce Device Management

Replace failed thin clients in **under 2 minutes**

Reduce time to **commission** and **configure** FactoryTalk Optix applications

Scalable Visualization

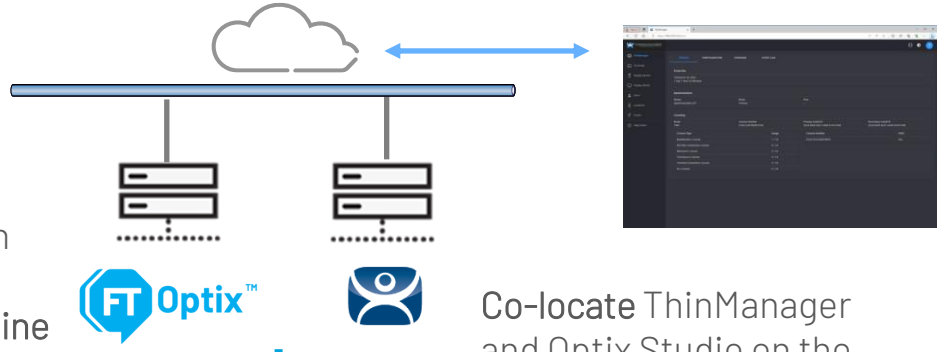
Promote HMI and terminal **configuration consistency** by configuring once, deploying to many

Deliver **other content** sources (Windows applications, IP cameras, VNC servers and more) with your ThinManager-managed FactoryTalk® Optix™ applications

FT Optix™ + THINMANAGER A ROCKWELL AUTOMATION TECHNOLOGY

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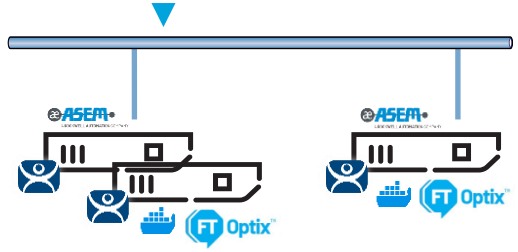
Proposed Architecture: ThinManager v14.2 + FactoryTalk® Optix v1.8 + ASEM 6300 (Simplified)



ThinManager Web UI: Shadow terminals from a web browser

Export .ZIP file to ThinManager Server: run **Optix runtime** utilizing **Native Presentation Engine**

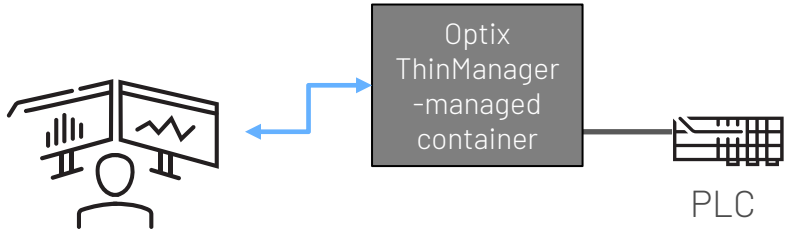
Co-locate ThinManager and Optix Studio on the same workstation



ASEM 6300 with storage and TPM to cache and restore last known firmware and configuration **without a network connection ****

*SSO requires ThinManager 14.2, Optix 1.6, and installed extension into Optix Studio

**Booting of container image still requires ThinManager Server connection with v14.2



Single sign on* into containerized **FactoryTalk® Optix™ application** from ThinManager terminal

Deploy multiple instances of FactoryTalk® Optix™ Optix applications to **provide redundant HMIs** (failover to another container vs. RDS session)

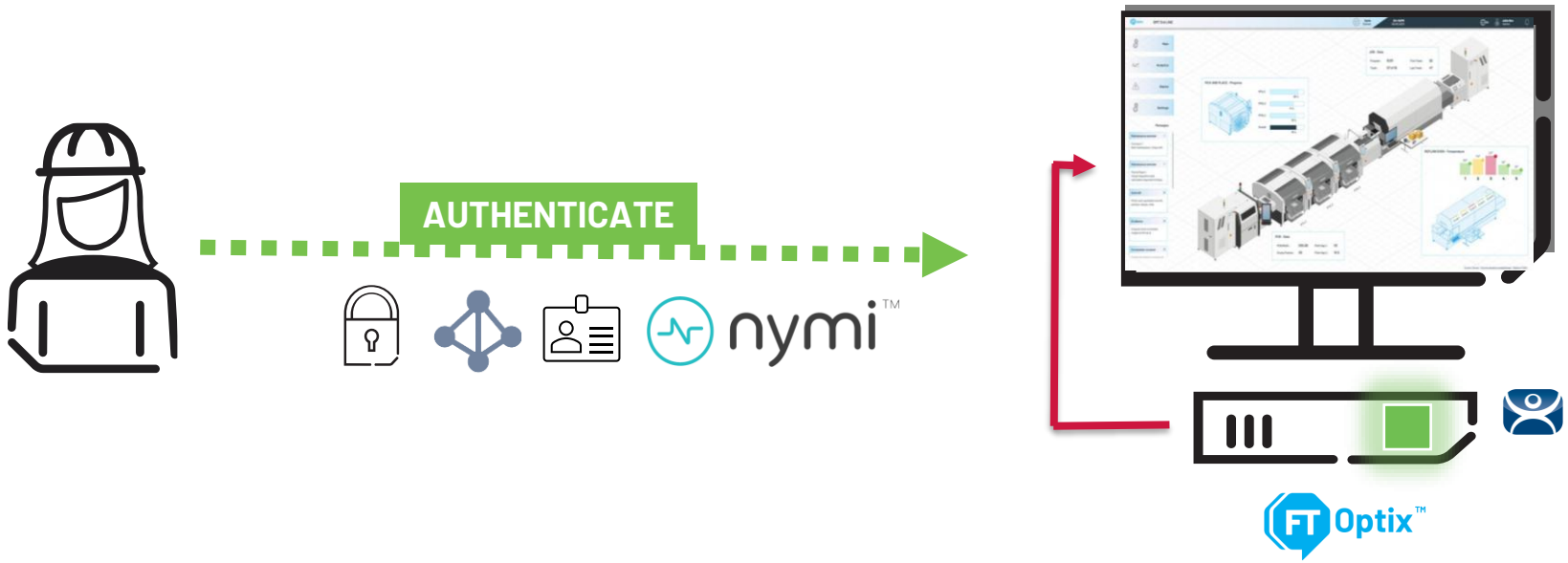
Coming soon!

Disconnected license support for Optix running in ThinManager container (Optix 1.8 and ThinManager 14.3)

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ThinManager user SSO into FactoryTalk Optix application

Pass ThinManager authentication through to FactoryTalk Optix applications running in a ThinManager container*



Increase operator productivity by **eliminating multiple login prompts** into both ThinManager and Optix separately

Leverage traditional ThinManager user authentication methods to **secure access to workstations and applications**

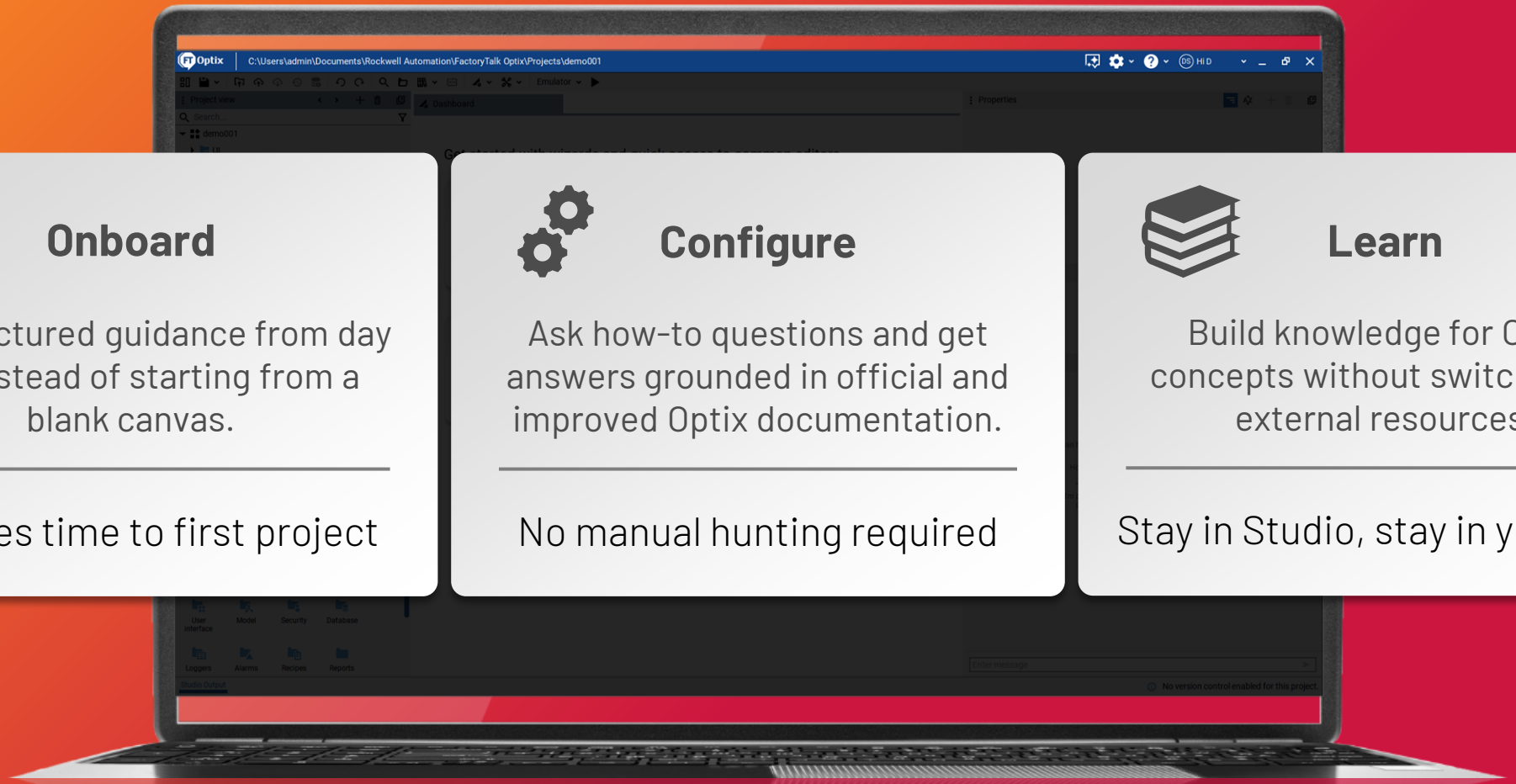
*Available with the release of FactoryTalk Optix 1.8. Requires **installation** of the FactoryTalk Optix ThinManager extension into Optix Studio (will be available on ThinManager.com following Optix 1.8 release)

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AI Design-Time Assistant

Your Product Documentation Expert

documentation-grounded guidance, reviewed by the user



Onboard

Get structured guidance from day one instead of starting from a blank canvas.

Reduces time to first project

Configure

Ask how-to questions and get answers grounded in official and improved Optix documentation.

No manual hunting required

Learn

Build knowledge for Optix concepts without switching to external resources.

Stay in Studio, stay in your flow

Embedded in FactoryTalk Optix Studio • Powered by approved documentation • Guided by you

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How It Works

Your product documentation expert

1 Ask in natural language

Type in the AI chatbot panel inside Studio. Ask like you'd ask a colleague: *"How do I configure an OPC UA connection?"*

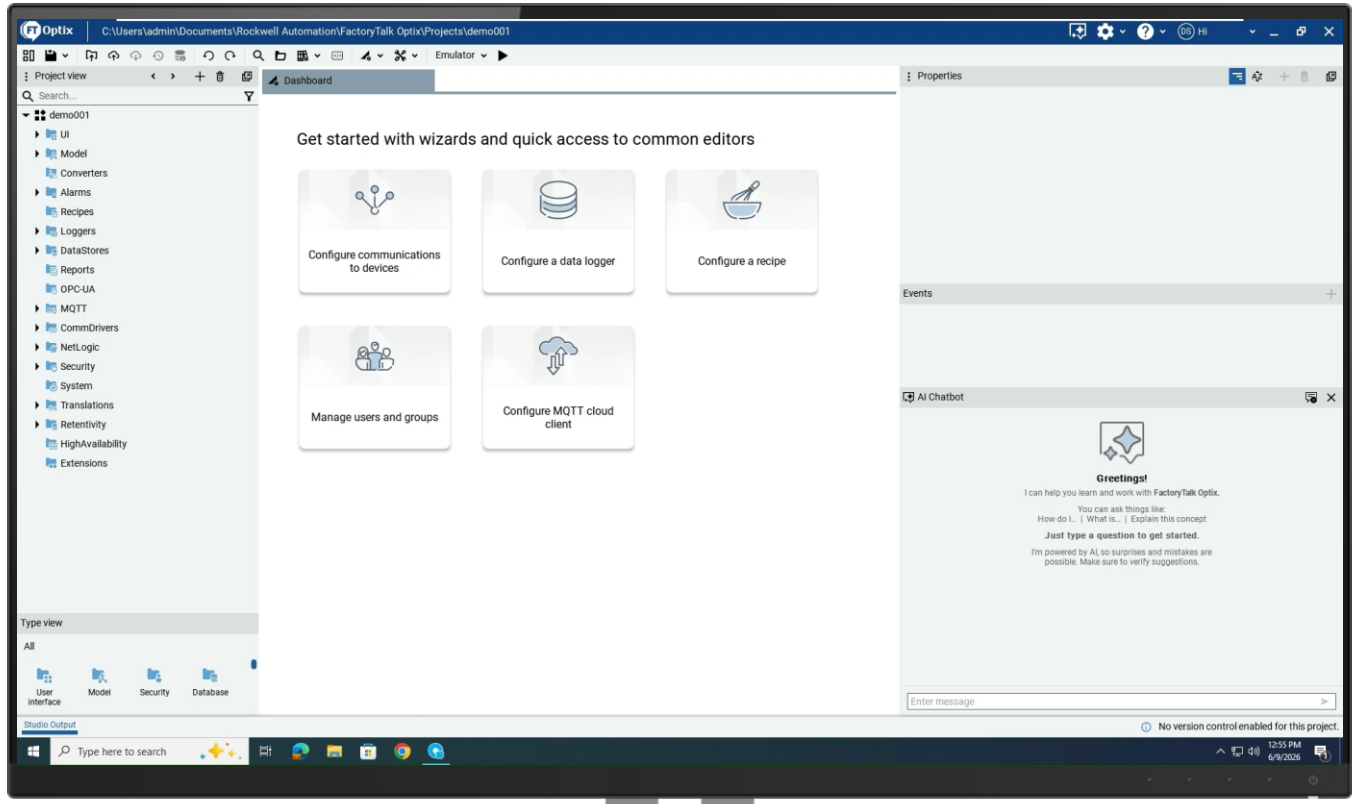
2 Optix docs are searched

Semantic search finds the most relevant content across official Optix documentation, understanding intent beyond keywords.

3 Answer appears in Optix Studio

A formatted response returns in the chat panel, structured steps, code blocks, and links to source documentation.

AI-generated answer based on Optix documentation. Verify critical steps before applying.



Project files stays local

The assistant sends the user's question + relevant documentation context to the LLM for inference.

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AI Design-Time Assistant: What's Included in Optix 1.8?

What you can do today with your product documentation expert

Integrated AI Assistant

Toolbar button opens a persistent right-side chat panel embedded in Optix Studio

Semantic Search

Understands your intent beyond keyword matching, surfacing related concepts and examples

Grounded in Approved Sources

Answers drawn from official Optix documentation only

Source Links

Jump directly to the source documentation behind any answer for full context

Copy & Reuse

Extract answers, steps, or code blocks straight into your project work

Session Context

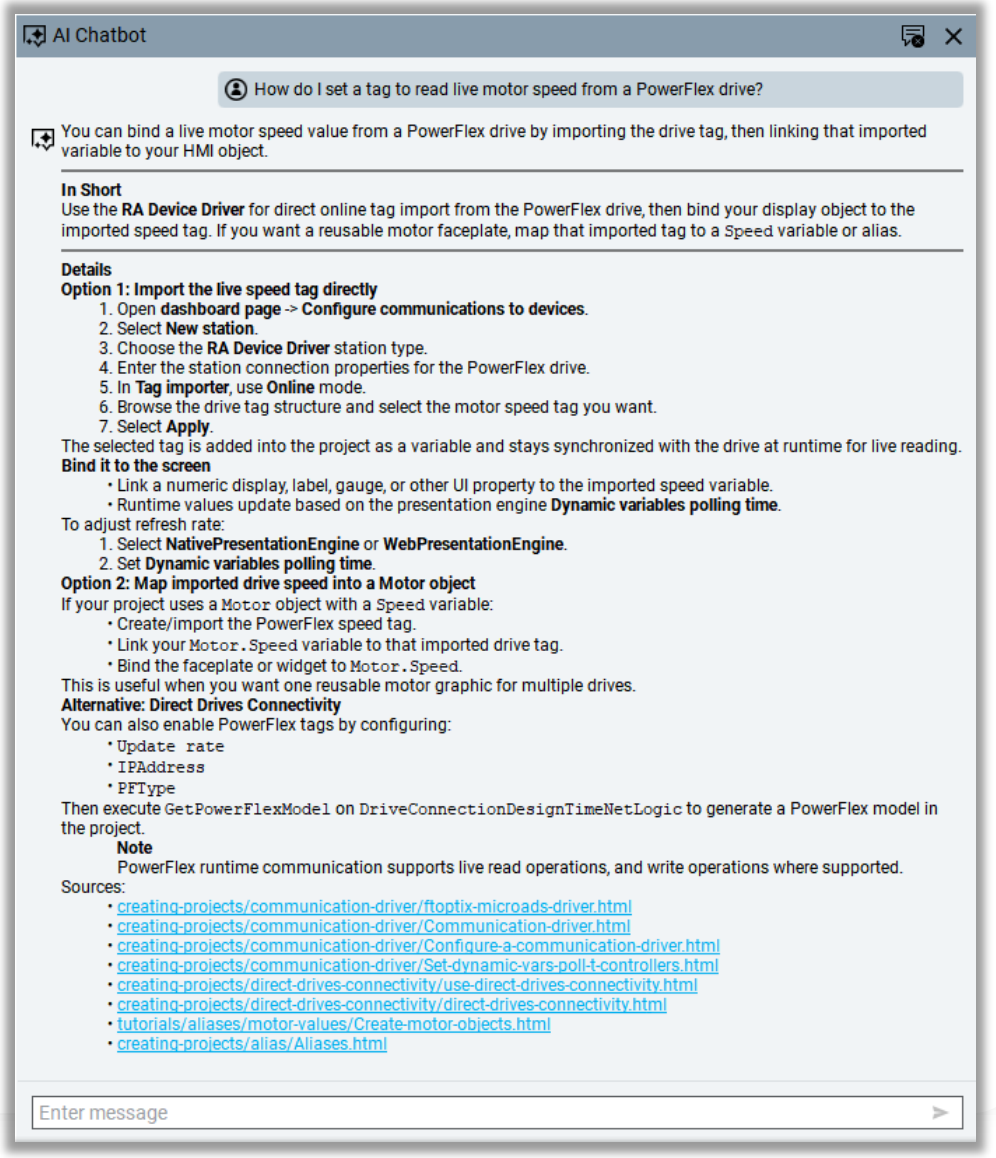
Carry context across follow-up questions within your session

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AI Design-Time Assistant: What It Does

What It Does

- ✓ Answers how-to, what-is, and concept questions about FT Optix
- ✓ Grounds every response in approved Optix documentation
  - *User guide, installation guide, release notes, NetLogic cheat sheet*
- ✓ Guided procedures with markdown formatting step-by-step answers
  - *Headings, code blocks, bold text*
- ✓ Carries conversation context within your project
  - *100 Q&A pair session memory*
- ✓ Links you to source documentation for any answer
- ✓ User stays in control by reviewing and verifying before acting



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Getting Started with the AI Design-Time Assistant

BEFORE YOU BEGIN

Desktop version of FT Optix Studio 1.8

installed and running

Pro subscription

active on your FT Hub organization

Internet access

required for AI responses

In 1.8, only the user's question + relevant docs are sent to the LLM.

YOUR FIRST SESSION

1

Open FT Optix Studio (desktop version)

Login to FT Hub, and Open the AI assistant chat panel

2

Ask a question in natural language

For example:

"How do I...?"

"What is...?"

"Explain this concept..."

3

Review the AI-generated response

Use provided guidance

Open referenced source content

Ask follow-up questions

Copy and reuse responses

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Responsible AI by Design

Trusted guidance, protected data, and clear operational boundaries

Trusted Answers

- Grounded in approved FT Optix documentation
- Includes links to supporting sources

User Stays in Control

- Advisory guidance only
- Users review and apply recommendations

IP & Data Protection

- Customer data is not used to train AI models
- Project files remain in FT Optix Studio
- Azure OpenAI provides inference services

Defined Boundaries

- No PLC writes
- No runtime control
- No autonomous project modifications

Built using Rockwell AI governance and Responsible AI practices.

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AI Design-Time Assistant Use Cases Examples

Scenarios where in-context guidance accelerates project development

ONBOARDING

Navigation + Screen Setup

"What's a good starter screen and navigation structure for a machine HMI?

CORE CONCEPTS

Types vs. Instances

"When should I create a custom type in Optix instead of copying objects?

SETUP GUIDANCE

Tag Binding + Connectivity

"What's the best practice to connect controller tags and bind them to widgets?

BEST PRACTICES

Data Logging + Trends

"How many data logs can I configure, and what's the recommended approach for trends in a typical machine application?

TROUBLESHOOTING

Configuration Symptoms

"My alarms aren't updating as expected, what should I check first in my project configuration?

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User Scenario: MQTT Configuration

Advanced how-to guidance helps experienced engineers configure faster

Optix User

How do I customize the MQTT payload in Optix, and where is that configured?

Pain Points:

- **Documentation hunting**

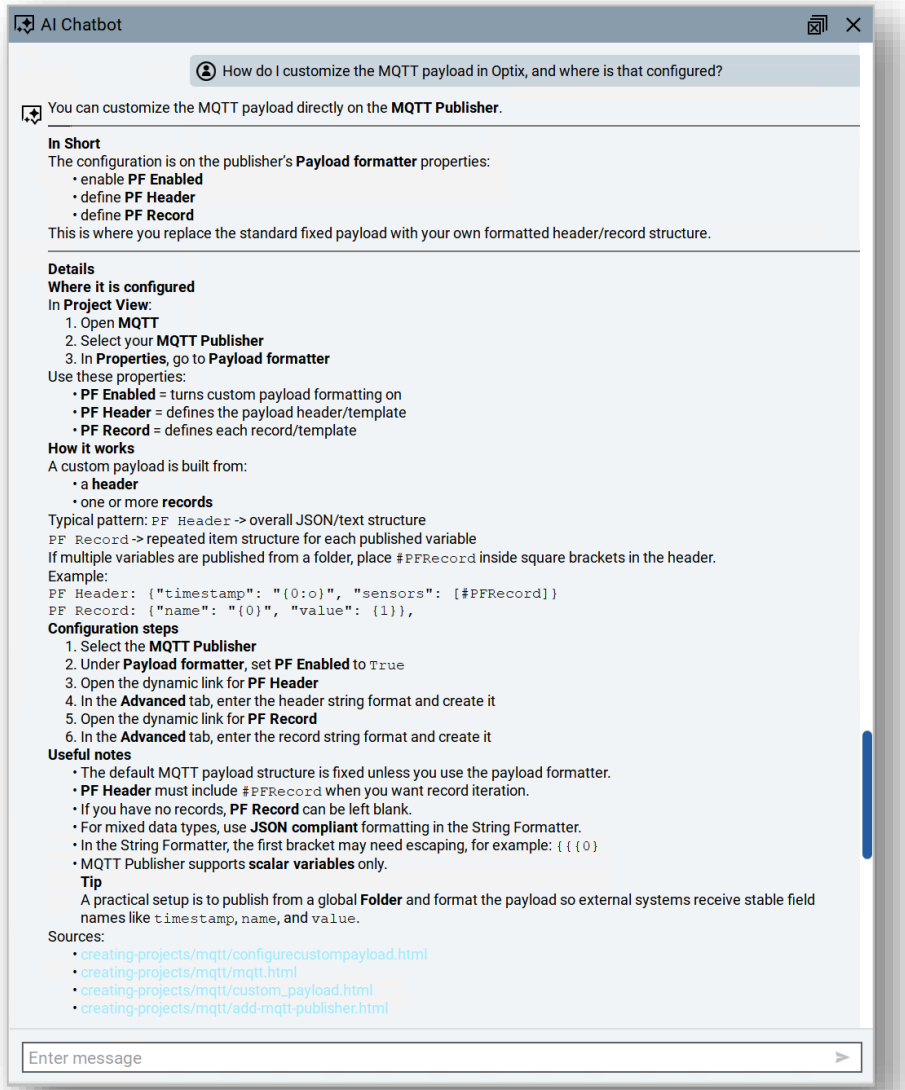
  Losing time searching documentation for specific configuration steps

- **Trial and error configuration**

  Without clear guidance on advanced settings, users spend time guessing and troubleshooting

- **Context-switching overhead**

  Switching between Studio and external docs breaks flow and slows down configuration tasks



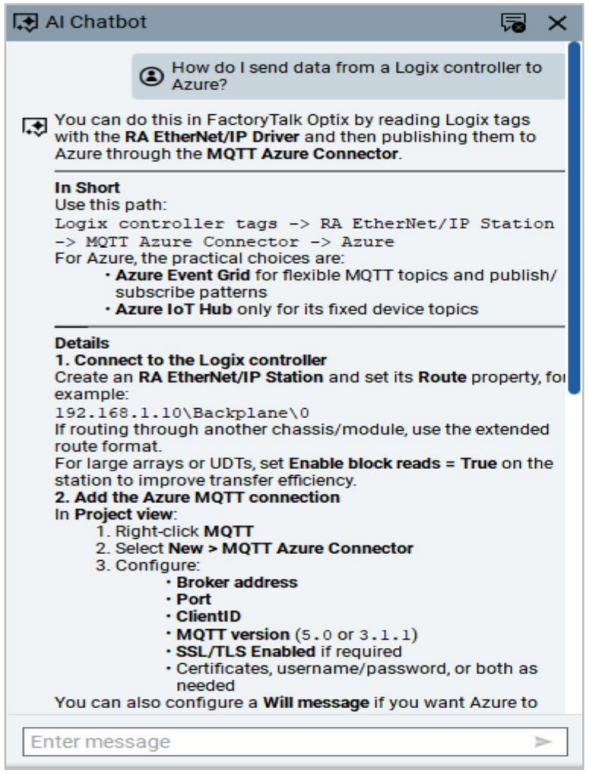
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FactoryTalk Optix AI Design Assistance

AI Design-Time Assistant: Foundational Release 1.8

Available in Optix 1.8

Create project content and find help through conversational text



Planned Future Direction

Generate screens from images

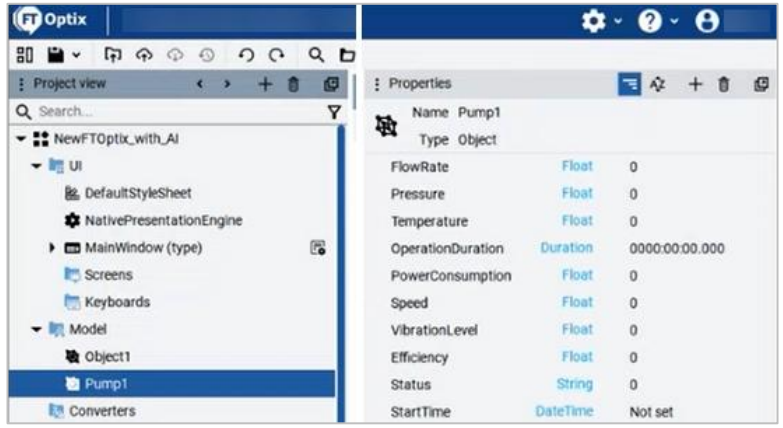
Figma, photo, hand-drawn sketch

Initial release planned in v1.9



Auto-generation of models, properties, & backend connectivity

Initial release planned in v2.0



FactoryTalk Optix Studio Pro entitlement required for all AI Design Assistance features

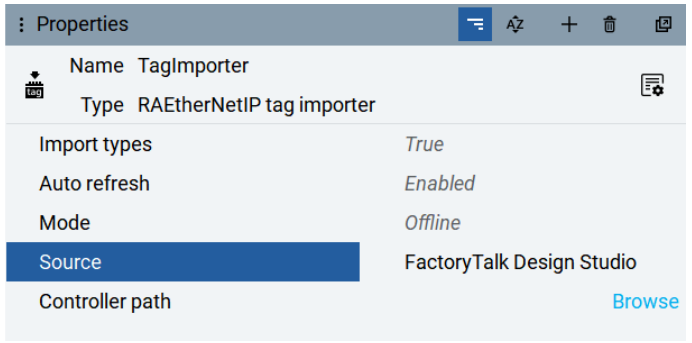
Subject to change:
Aug 2026

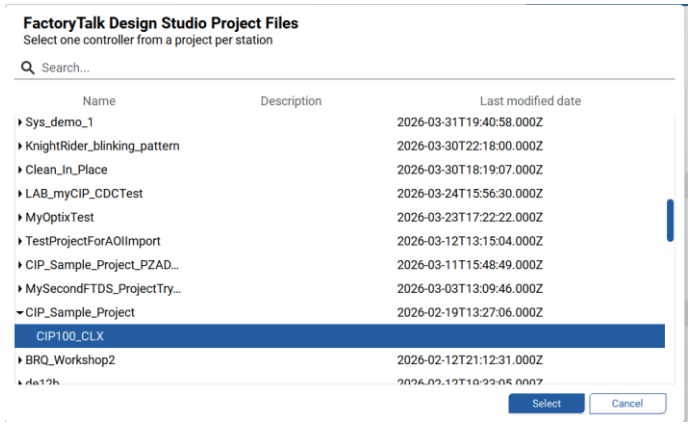
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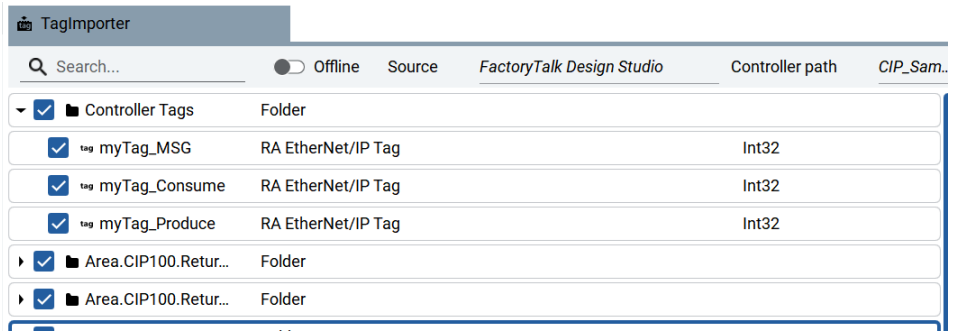
FactoryTalk Design Studio Support

Offline tag import

- Import tags offline from a Logix controller configured in an FactoryTalk Design Studio project
- Enables the ability to import tags prior to deploying the FactoryTalk Design Studio application to a Logix controller
- Utilizes new offline tag import process for FactoryTalk Design Studio projects:
  1. Choose Offline mode
  2. Select FactoryTalk Design Studio source
  3. Browse to FactoryTalk Design Studio project and select a controller
  4. Import the required tags
- Support for FTDS device names and routine-scoped tags







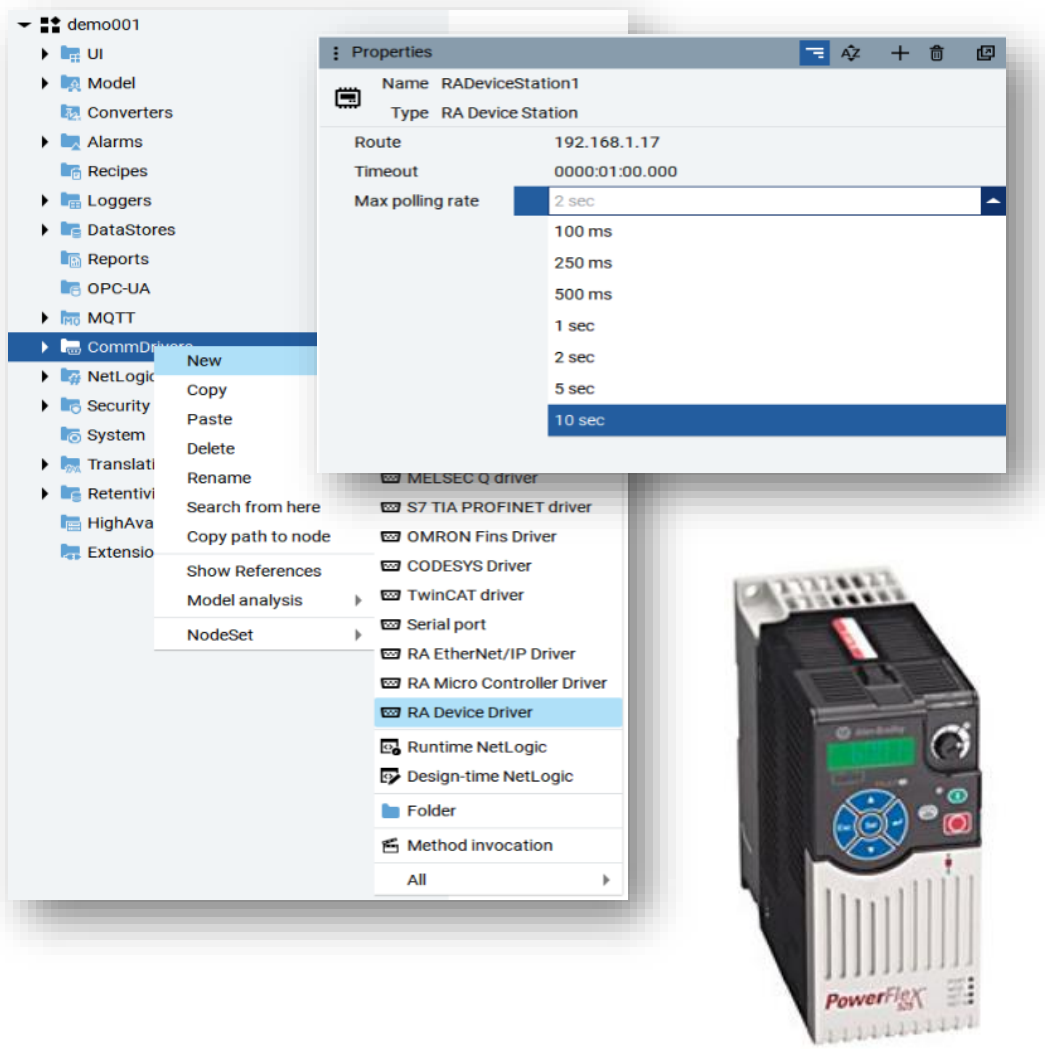
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Data Connectivity

RA Device Driver

Direct connectivity to PowerFlex drives

- Enables direct communication with PowerFlex drives over EtherNet/IP
- Provides DPI Port 0 parameter access
- Validated with:
  - *PowerFlex 525 (PF525)*
  - *PowerFlex 755 (PF755), including these variants:*
    - *PF755TS*
    - *PF755TR*
    - *PF755TL*
    - *PF755 TM Bus Supply*
    - *PF755 TM-CBI*
  - *PowerFlex 6000T (PF6000T)*
  - *PowerFlex 7000 (PF7000)*

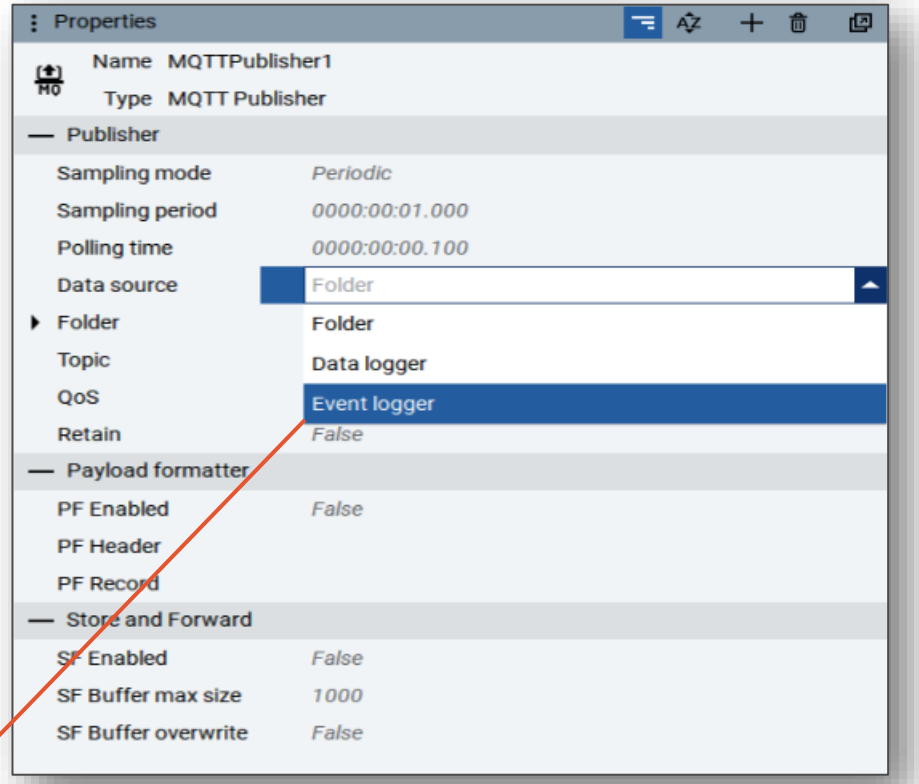


Publish Event Logger data directly to an MQTT broker

Data Connectivity

Publish Event Logger data through the MQTT Client by mapping an existing, user-defined Event Logger to an MQTT Publisher.

- Users can now stream runtime event data to an MQTT broker so external systems, supervisory applications, or cloud tools can consume event information.
  - Publish and monitor events across a network.
  - Bridge local system monitoring with cloud or remote analytics.
  - Reduce reliance on manual data extraction or custom middleware.



- Enable real-time remote monitoring
- Simplify integration with cloud analytics
- Leverage existing Event Logger configurations

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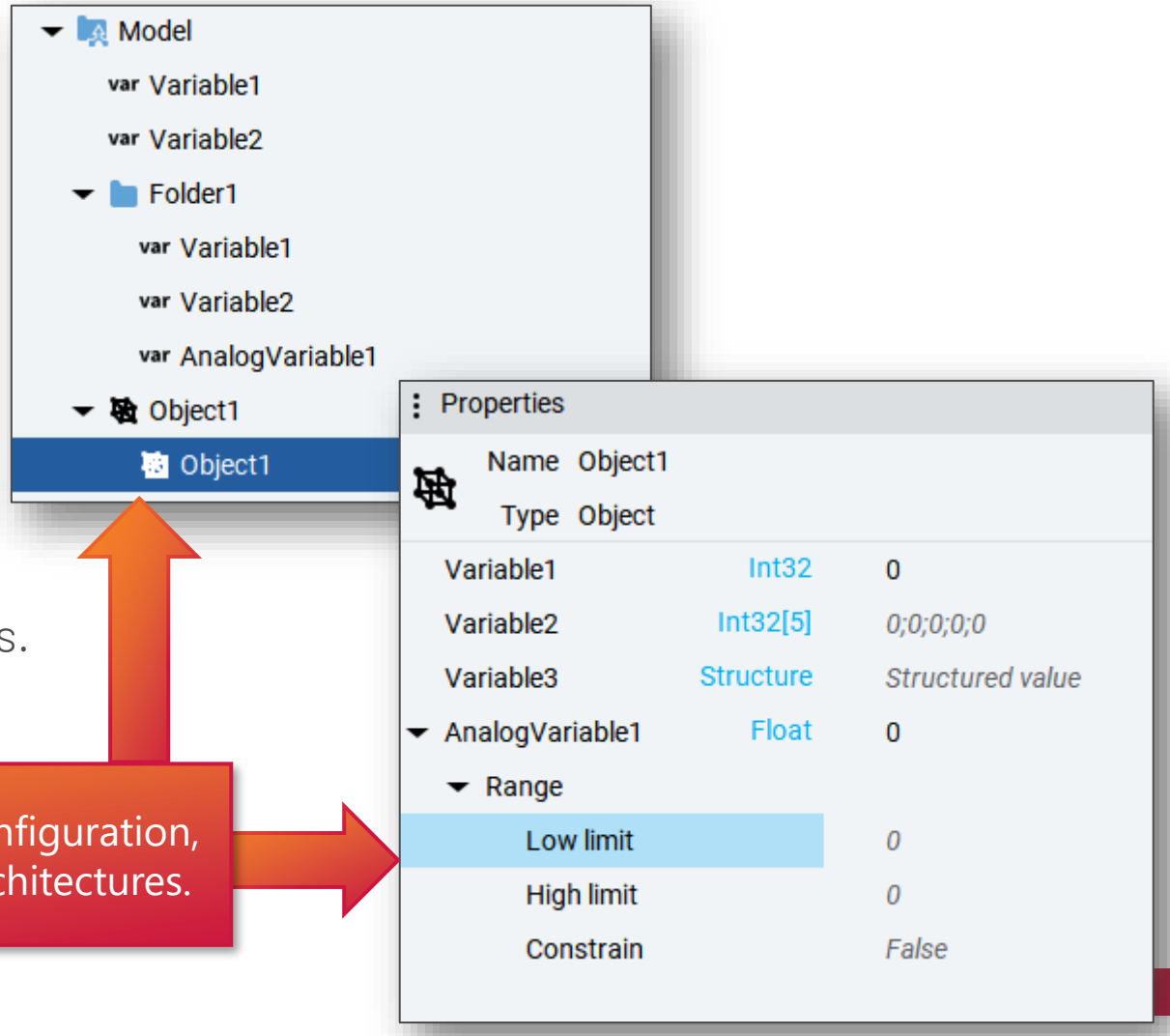
MQTT Client: Complex Data Types

Data Connectivity

With FT Optix 1.8 MQTT communication is no longer limited to simple atomic values. The MQTT Client module (both Publisher & Subscriber) now supports:

- Folders and subfolders
- Objects and nested objects
- Variables and sub-variables
- Properties
- Arrays
- Complex structures

This enhancement allows entire data models to be exchanged through MQTT, reducing engineering effort and improving scalability for Industrial IoT and edge applications.



This enables richer data exchange, simplified project configuration, and seamless integration with modern Industrial IoT architectures.

Extended Runtime Licensing

What Extended Licensing Means for Customers

Extended Runtime Licensing gives customers flexibility today and confidence their Optix application can scale tomorrow

Pay Only for What You Need

Select only the runtime features you actually use. Feature Tokens scale with your application needs without counting tags, alarms or screens.

Scale from Simple to Advanced

The same licensing model supports basic applications through advanced applications. You can start with a Base Runtime package and add an Extended Runtime package later.

Clear Path to High Availability

Enable multi-node High Availability in FactoryTalk Optix 1.8 on any existing installable Optix Runtime software. Add one Extended Runtime Package to support a full 3-node HA cluster, not three separate runtime licenses.

Transparent, Predictable Sizing

Runtime size is driven by the features you use. The Runtime Sizing Calculator aligns licensing directly to your application requirements.

Future-Ready Investment

Extended Licensing aligns with the "pay as you grow" model. Designed to evolve as your applications expand or adopt more advanced runtime capabilities.

Flexibility today, confidence to scale tomorrow

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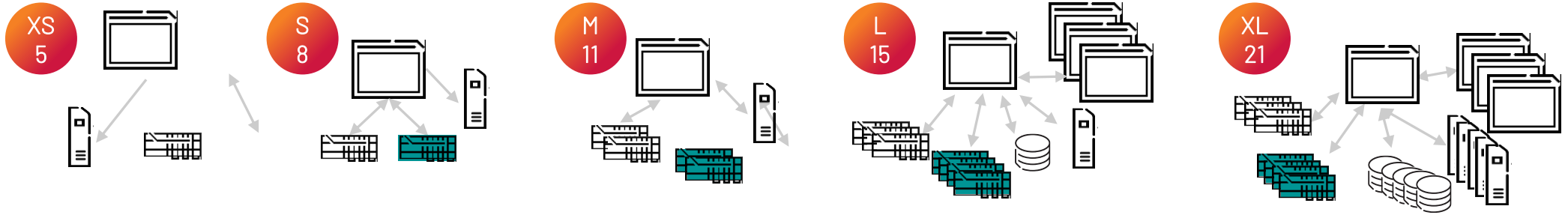
Today - FactoryTalk® Optix™ Base Runtime Licensing

Scalable runtime licenses aligned to an application's specific requirements

Smaller applications with focus on device data management or simple feature set

Medium-sized applications with more connections, richer features

Larger, feature-rich applications



Pay only for what you need

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New - FactoryTalk Optix Base + Extended Runtime Licensing

Available with FactoryTalk Optix v1.8 (AFC Early Fall 2026)

Base Feature Token Packages	XS	S	M	L	XL
Base Feature Tokens	5	8	11	15	21

1
Start with a Base Token Package

Extended Feature Token Packages	XXS	XS	S
Extended Feature Tokens	3	5	10

2
Add Extended Token Package when needed

Pay only for what you need

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Summary of features requiring Base Runtime Tokens

Please check the runtime sizing tool for a complete list of features and capabilities

General Features	Base Feature Tokens
Native Presentation Engine	1
Web Clients – 1 session	1
Web Clients – up to 3 sessions	2
Web Clients – up to 5 sessions	3
Alarms + Alarm Retentivity + Alarm Event Logging	1
Recipe	1
Active Directory Authentication	1
Basic Reporting	1
Audit Signature	1
Runtime Retentivity	1
Event Logger	1
Data logger	1

Database & Controller Connectivity	Base Feature Tokens
External Database – 1 connection [1]	1
External Database – up to 3 connections [1]	2
External Database – up to 5 connections [1]	3
Store and Forward	1
Embedded InfluxDB Database	1
Embedded SQLlite Database	0
Single controller Connection [1]	1
Multiple Controller Connections [1]	2
Runtime tag upload [1]	1
Rockwell Hardware [1]	-1

Interoperability Features	Base Feature Tokens
OPC UA client - Single	1
OPC UA client – Multiple	2
OPC UA server – 1 connection	1
OPC UA server – up to 3 connections	2
OPC UA server – up to 5 connections	3
MQTT Broker – up to 25 clients	1
MQTT Client (Subscriber) [2]	1 per third-party broker connection
MQTT Client (Publisher) [2]	1 per third-party broker connection

[1] *For more details, check the runtime sizing tool*

[2] *Optix embedded MQTT broker connections do not require tokens*

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Features requiring Extended Runtime Tokens

Introduced with FactoryTalk Optix v1.8

Advanced Features	Extended Feature Tokens
Three-Node High Available Cluster	3
Workstation Clients – 1 session	1
Workstation Clients – up to 3 sessions	2
Workstation Clients – up to 5 sessions	3
Workstation Clients – up to 10 sessions	5

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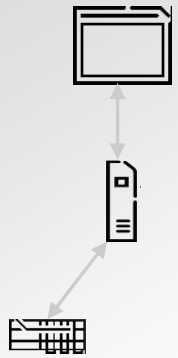
Optix Runtime with Base Runtime Package

Example: Optix Runtime with Base Runtime package (XS)

This example architecture has the following

- Built-in touch screen is not required
- Communicates with a Rockwell Automation controller
- Acts as an OPC UA server
- Logs data to an internal database
- One web client

System architecture – Extra Small



Runtime package options

- Optix Base Runtime package (XS)
- or Optix Base Runtime package (S) for future expansion of basic features

Deployment options

Hardware with Optix Embedded

- Embedded Edge Compute
- OptixEdge Standard

Installable Optix Runtime software

- Optix Runtime Base Package (XS)
- Consider bundling with ASEM 6300 PCs

Notes

- XS Base Runtime includes 5 base tokens; S Base Runtime includes 8 base tokens.
- There is no need for an extended package.
- Use the Device Sizing tool to select the best deployment option for your application.

Runtime Sizing Calculator

	Base Tokens	Extended Tokens
Clients Web HMI Client - 1 connection	1	
Features Runtime retentivity Data logger Embedded Database	1 1 0	
Controller Connections RA EtherNet/IP - 1 connection	1	
Interoperability OPC UA Server - 1 connection	1	
High Availability None	-	
Total	5	-

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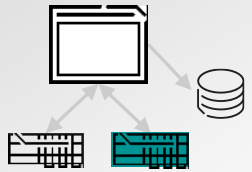
Optix Runtime with Base Token Package

Example: Optix Runtime with Base Runtime package (S)

This example architecture has the following

- Single HMI runtime with a touch screen at least 10 inches (wide screen)
- Typical HMI functionality
- Communicates with a Rockwell Automation controller and a third-party controller
- Logs data to an internal database
- Requires event logging, alarming, and recipe

System architecture - Small



Runtime package options

- Optix Base Runtime package (S)
- or Optix Base Runtime package (M) for future expansion; leaves 3 base feature tokens available for future use of basic features.

Deployment options

Hardware with Optix Embedded

- OptixPanel Standard

Installable Optix Runtime software

- Optix Runtime Base Package (S)
- Consider bundling with ASEM 6300 Panel PCs

Notes

- S Base Runtime includes 8 base tokens.
- There is no need for an extended package.
- Base Runtime (M) includes 11 base tokens. 3 base feature tokens will be available for future use of basic features.
- Use the Device Sizing tool to select the best deployment option for your application.

Runtime Sizing Calculator

	Base Tokens	Extended Tokens
Clients Native HMI Client	1	
Features Alarms Recipe Runtime retentivity Event Logger Data logger Embedded Database	1 1 1 1 1 0	
Controller Connections RA EtherNet/IP - 1 connection Third-party drivers - 1 connection	1 1	
Interoperability None		
High Availability None		
Total	8	-

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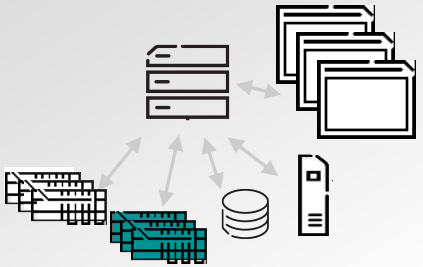
Optix Runtime with Base & Extended Token Packages

Example: Optix Runtime with Base Runtime package (L) and Extended Runtime package (XXS)

This example architecture has the following

- An HMI application with typical HMI functionality with five web clients
- High Availability (HA)
- Communicates with multiple Rockwell Automation controllers and multiple third-party controllers
- Acts as a client to other OPC UA servers
- Logs data to an external database
- Requires event logging, alarming, recipe, and basic PDF reporting

System architecture – Large w/High Availability



Runtime package options

- Optix Base Runtime package (L) + Extended Runtime package (XXS)
- Or Optix Base Runtime package (XL) + Extended Runtime package (XS) for future expansion of basic and advanced features.

Deployment options

Installable Optix Runtime software

- Optix Base Runtime package (L) + Extended Runtime package (XXS)
- consider bundling with ASEM 6300 PCs

Notes

- Base Runtime (L) includes 15 base tokens.
- Extended Runtime package (XXS) includes 3 extended tokens.
- Extended Runtime package (XS) includes 5 extended tokens.
- Use the Device Sizing tool to select the best deployment option for your application.

Runtime Sizing Calculator

	Base Tokens	Extended Tokens
Clients Web HMI Clients - 5 connections	3	
Features Alarms Recipe Data logger Basic Reporting Event Logger Runtime retentivity External Database	 1 1 1 1 1 1 1	
Controller Connections RA EtherNet/IP - 4 connections Third-party drivers – 2 connections	 2 2	
Interoperability OPC UA Client - 1 connection	1	
High Availability Three-node HA Cluster		3
Total	15	3

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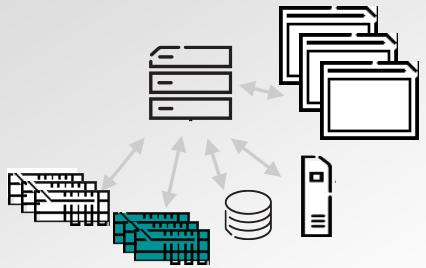
Optix Runtime with Base & Extended Token Packages

Example: Optix Runtime with Base Runtime package (XL) and Extended Runtime package (S)

This example architecture has the following

- An HMI application with typical HMI functionality
- Ten workstation clients and five web clients
- High Availability (HA)
- Communicates with multiple Rockwell Automation controllers and multiple third-party controllers
- Acts as a client to other OPC UA servers
- Logs data to an external database
- Requires event logging, alarming, recipe, and basic PDF reporting

System architecture – HA & Workstation Clients



Runtime packages

- Optix Runtime Base Package (XL) + Extended Package (S)
- 3 base feature tokens and 2 extended feature tokens are available for future use.

Deployment Options

Installable Optix Runtime software

- Optix Runtime Base Package (L) + Extended Package (S)
- Consider bundling with ASEM 6300 PCs

Notes

- Base package (XL) includes 21 base tokens.
- Extended package (S) includes 10 extended tokens.
- In this example, 3 base feature tokens and 2 extended feature tokens are available for future use.
- Use the Device Sizing tool to select the best deployment option for your application

Runtime Sizing Calculator

	Base Tokens	Extended Tokens
Clients Web HMI Clients - 5 connections Workstation Clients – 10 connections	3	5
Features Alarms Recipe Active Directory Authentication Event Logger Data logger Basic Reporting Audit signature capture Runtime retentivity External Database	 1 1 1 1 1 1 1 1 1	
Controller Connections RA EtherNet/IP - 4 connections Third-party drivers – 2 connections	 2 2	
Interoperability OPC UA Client - 2 connections	2	
High Availability Three-node HA Cluster		3
Total	18	8

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Tools and content to help with runtime licensing

Determining what license is required for your application

- Updated Runtime Sizing Calculator will release with V1.8
- FT Optix Studio will tell you what size license the current application requires, including Extended Runtime licenses
- Optix Ordering Guide updated to cover Extended Runtime licenses

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Firmware 7.0.0.x for Optix-based devices

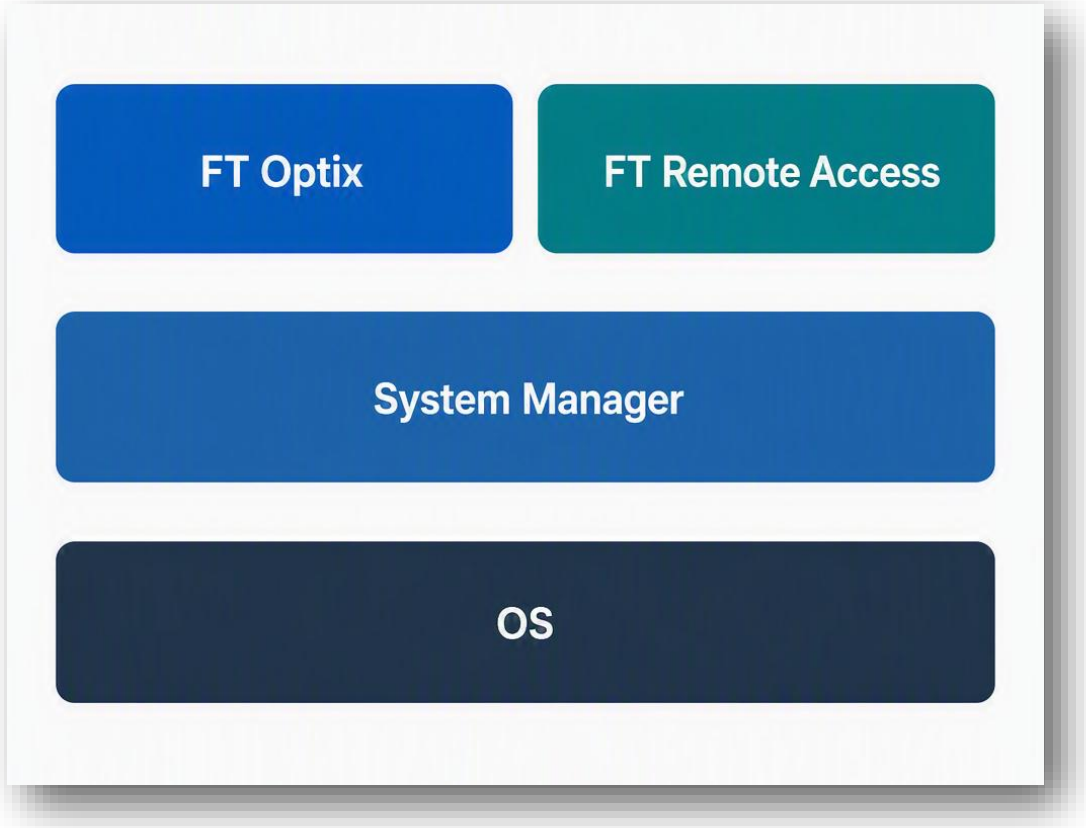
Software Components Included

Firmware 7.0.0.x

Software Stack

The 7.0.0.x firmware for OptixPanels, OptixEdge and Embedded Edge Compute includes:

- FactoryTalk® Optix 1.8 (minimum version)
- FactoryTalk® Remote Access 14.3.2
- System Manager 1.8
- Linux Yocto Scarthgap with Kernel 6.12



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Local Web-based Remote Desktop

Firmware 7.0.0.x

Feature

It is now possible to access the HMI screen directly using a Web Browser over the local network allowing to view and mirror the HMI screen on smartphones, tablets, and Linux PCs. Local Connection option must be enabled in System Manager.



Benefits

- Enables local, on-demand access without installing FactoryTalk® Remote Access™ Tools.
- Greater flexibility for operations and maintenance across OptixPanel.

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Access to USB Memory and SD Card via File Transfer

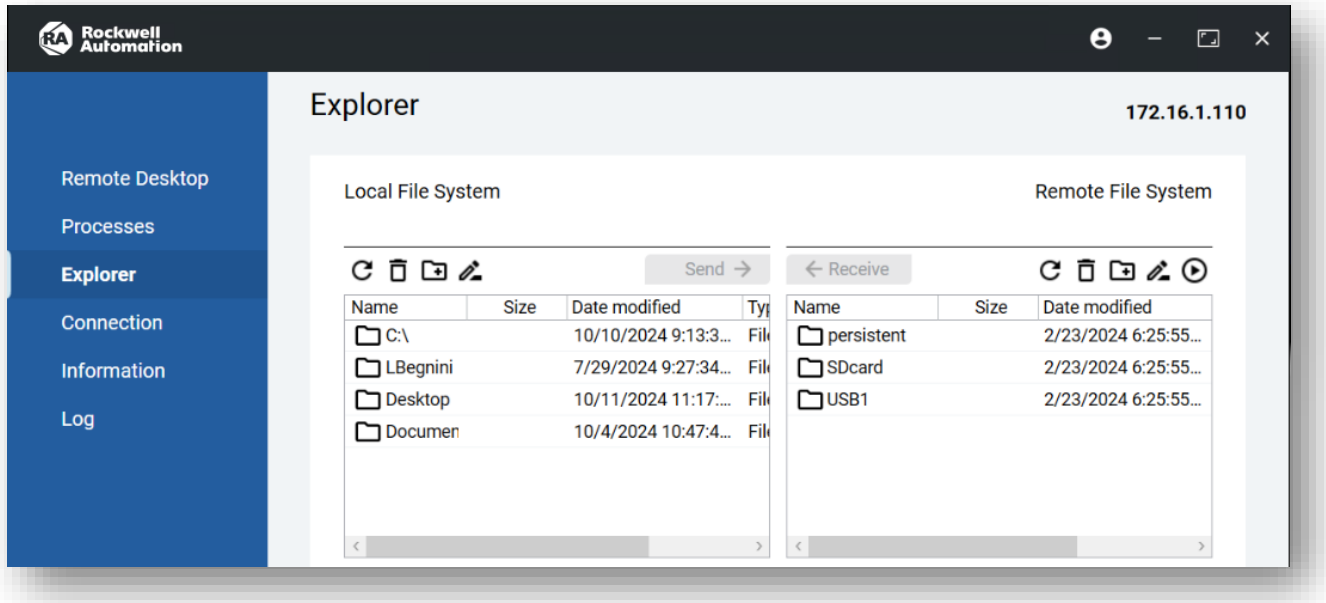
Firmware 7.0.0.x

Feature

It is now possible to remotely access USB memory devices and SD cards plugged into the device through the File Transfer feature available in the FactoryTalk® Remote Access™ Tools.

Benefits

Enabling remote file management on external storage media for OptixPanels, OptixEdge, Embedded Edge Compute and Stratix 4300.



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Show Remote Access info in System Manager

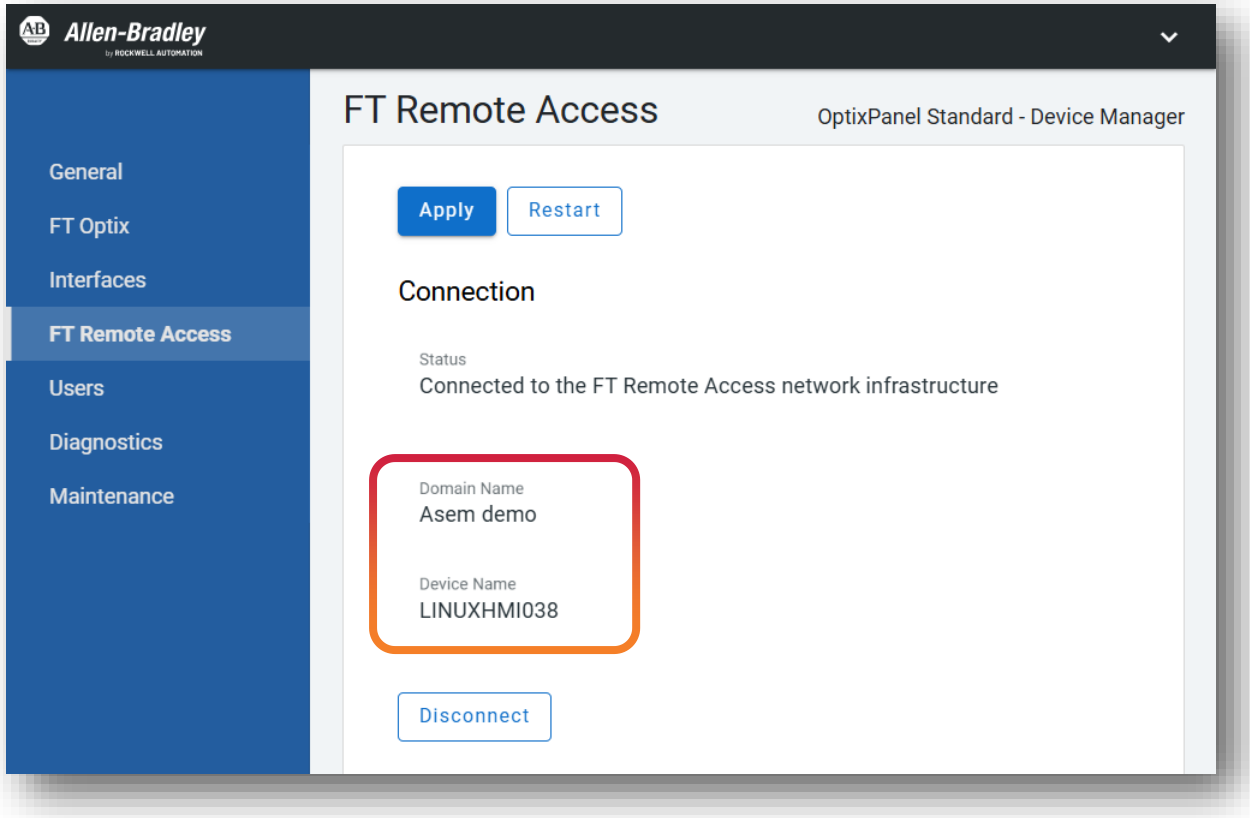
Firmware 7.0.0.x

Feature

System Manager now displays the domain name and the device name assigned in the FactoryTalk® Remote Access™ domain. This information is available under the FT Remote Access section.

Benefits

Improving device/domain identification for OptixPanels, OptixEdge, Embedded Edge Compute and Stratix 4300



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Bundle image support

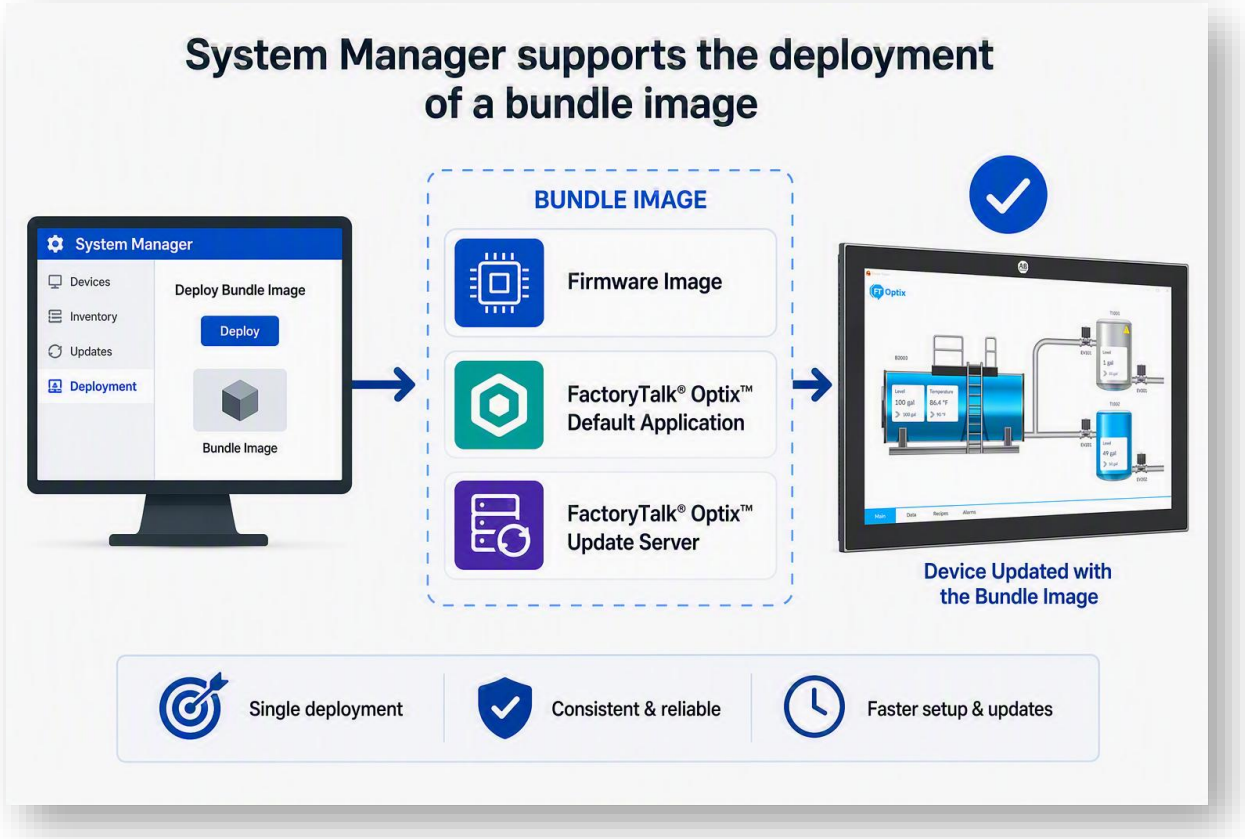
Firmware 7.0.0.x

Feature

System Manager now supports the deployment of a bundle image that includes the Firmware image, the FactoryTalk® Optix™ Update Server and the Default Application.
Available for OptixPanels, OptixEdge and Embedded Edge Compute.

Benefits

- Possibility to upgrade*
  the entire software stack of the device
- Possibility to perform downgrades*,
  which is useful for customers who need to run previous software versions on newly delivered devices that ship with the latest packages.



* The upgrade/downgrade with bundle image overrides the FT Optix application deployed on the device.

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Device Backup and Restore

Firmware 7.0.0.x

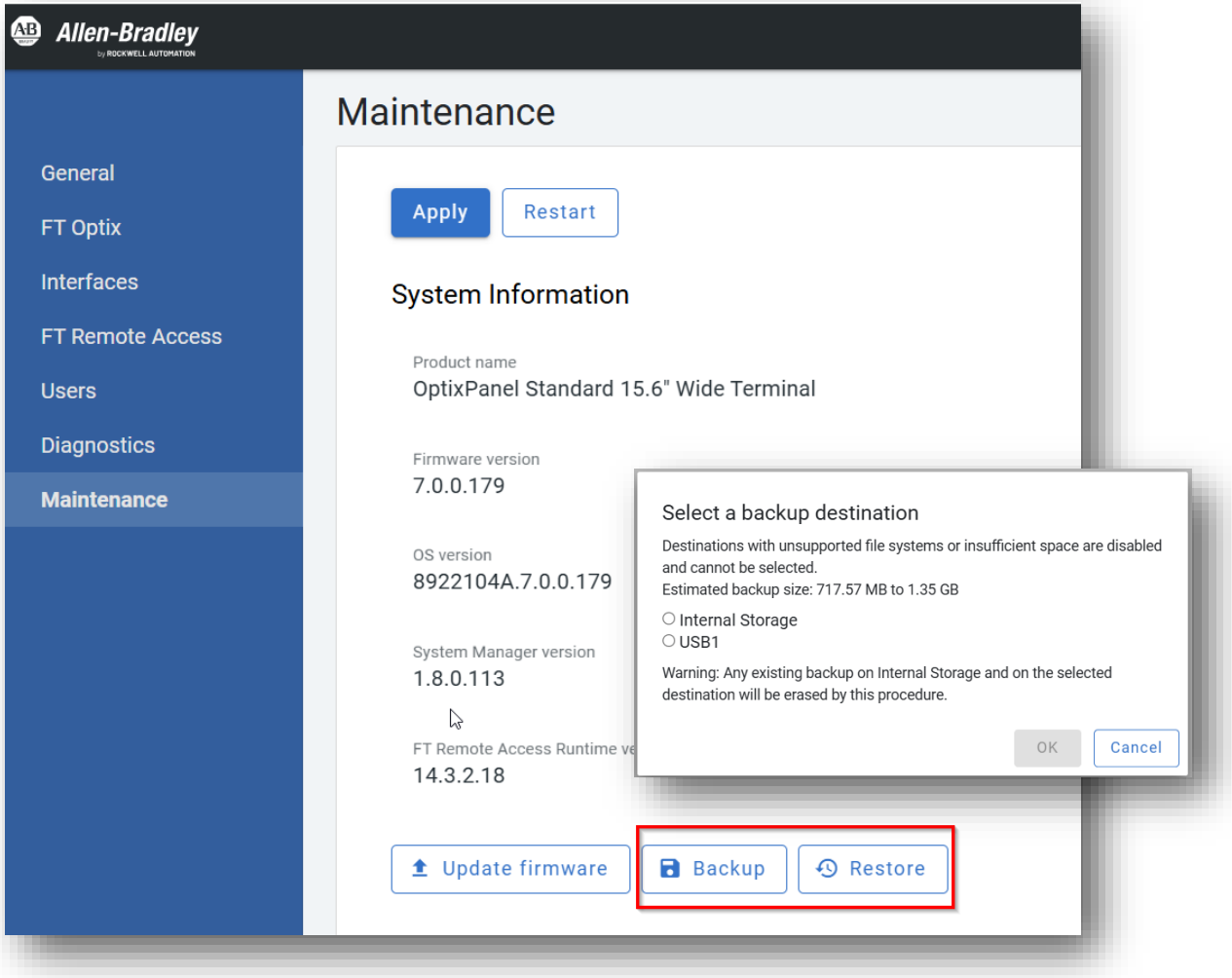
Feature

System Manager now includes a Backup and Restore feature that provides a mechanism to capture and reuse the state of a device.

Available for OptixPanels, OptixEdge and Embedded Edge Compute.

Benefits

- The backup can be used to restore the same device to a previous known state
- Or to apply the same system configuration to another device of the same type when the original device is no longer recoverable.



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OptixEdge Advanced networking

Firmware 7.0.0.x

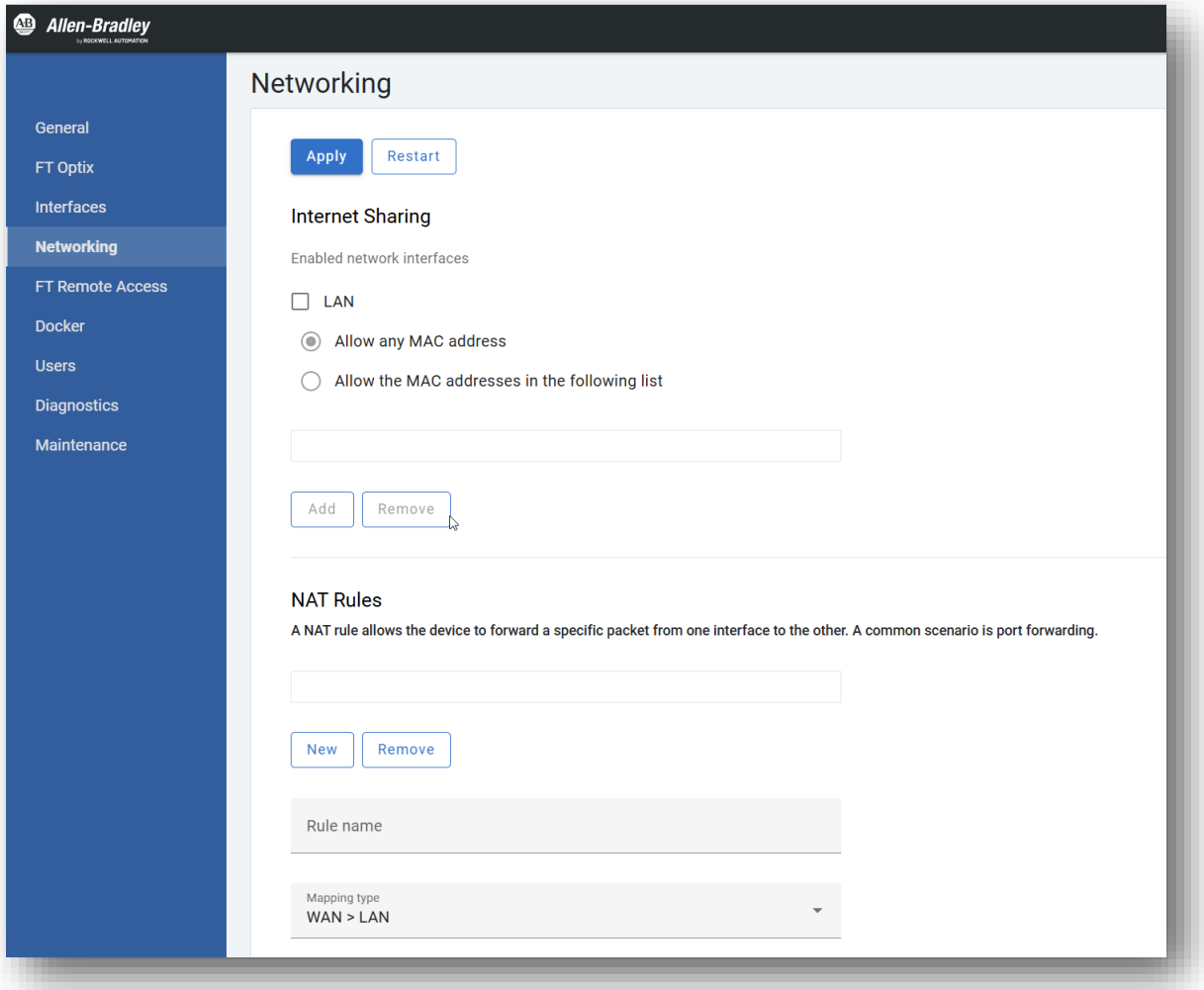
Feature

OptixEdge now includes the same advanced networking capabilities available on Stratix 4300, including Internet Sharing, NAT Rules and Routing Rules that can be configured via System Manager web interface.

Benefits

This enables the device:

- to share Internet connectivity with automation subnetworks,
- define NAT mappings between WAN and LAN interfaces,
- configure static Routing rules for communication across different networks



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USB Device Configurator

Firmware 7.0.0.x

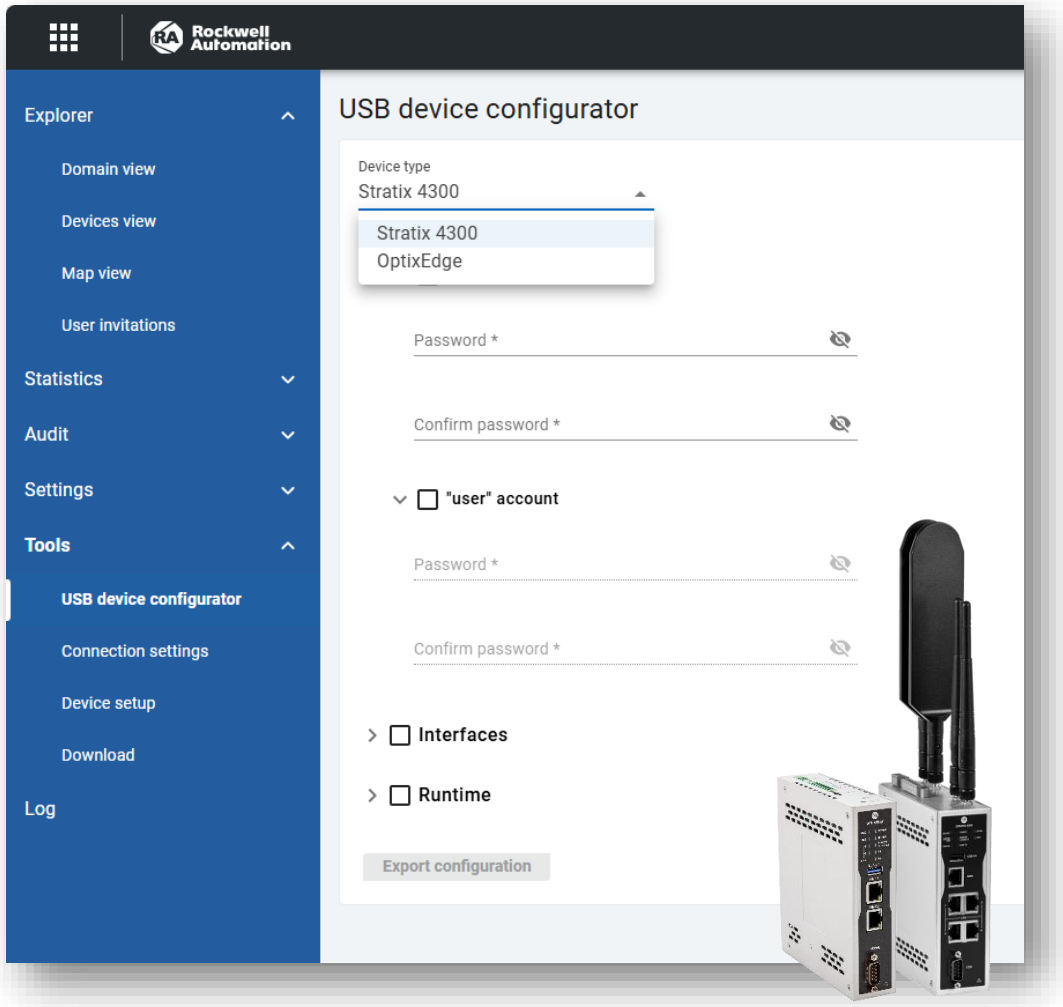
Feature

Enable device configuration and registration to FactoryTalk® Remote Access™ domain through USB stick via JSON configuration file that can be crated by FactoryTalk® Remote Access™ Manager.

Supported Devices: Stratix 4300 and OptixEdge

Benefits

- Consistent, repeatable configuration using a standardized JSON file
- Reduced setup errors and effort for field and commissioning teams



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2026 FactoryTalk Optix webinar series

2026 Webinar Series: *Visualization Deep Dive*

Live and on-demand to fit your schedule

Register online for the series

Visualization Deep Dive Series: HMI with modern architectures and AI

A three-part series exploring modern, scalable SCADA architectures using FactoryTalk Optix, including multi-node and high-availability deployments leveraging purpose-built hardware platforms. The series concludes with how AI-driven visualization with FactoryTalk View and FactoryTalk Optix helps teams deliver smarter operator experiences.



August 6, 2026 - 10:00 AM CST

Resilient OT operations with IT technology

Learn how FactoryTalk Optix enables resilient operations with 3-node high availability, leveraging modern technology with containers and Kubernetes.

September 15, 2026 - 10:00 AM CST

FactoryTalk Optix Multi-Node Architectures: Integrated Hardware and Software

FactoryTalk Optix works together with Rockwell Automation HMI hardware to provide a purpose-built edge deployment foundation.

October 28, 2026 - 10:00 AM CST

Smarter Visualization: Introducing AI Assistants with FactoryTalk Optix and FactoryTalk View

See how FactoryTalk Optix and FactoryTalk View integrate AI assistants into visualization design and runtime workflows, through real OEM and SI use cases.

All webinars will be available on demand following the live session

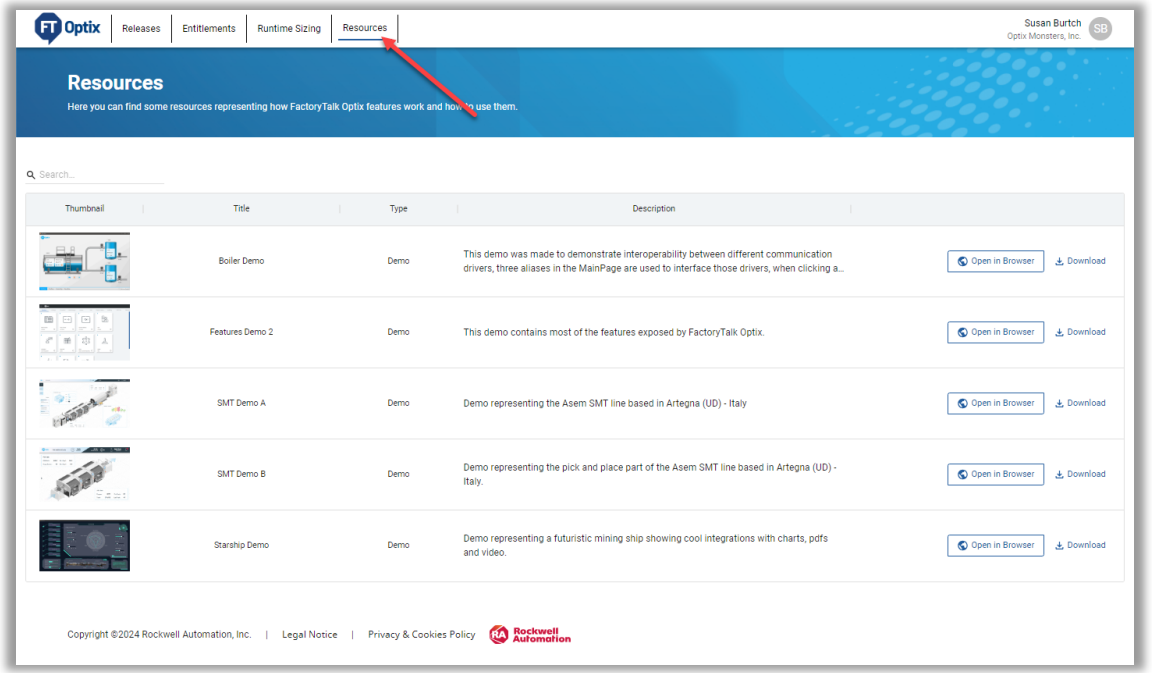
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Online Demos

FactoryTalk Optix Instant Online Customer Demos!

Online demos

- Resources tab provides access to many FactoryTalk Optix demo applications
  - These demo applications can be viewed using your browser
  - A description for each demo is also available
  - Use the Search bar to look for a specific demo
- Available to anyone with a FactoryTalk Hub account, including customers!
- Click **Open in Browser** to run each demo application in your web browser
  - Boiler demo
  - Features demo
  - ASEM factory machines demo
  - Trade show and event demo applications
  - More coming soon!



Now available: Preview demo application in browser, then download demo application files from GitHub on the Resources tab

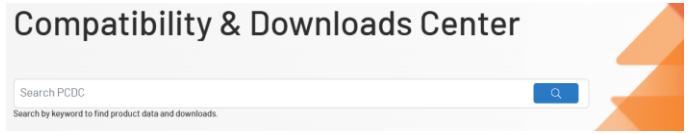
PCDC Release Notifications

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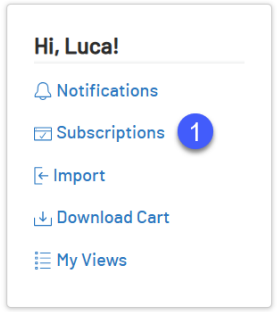
How to get notification of a new release on PCDC

PCDC allows you to subscribe to new releases

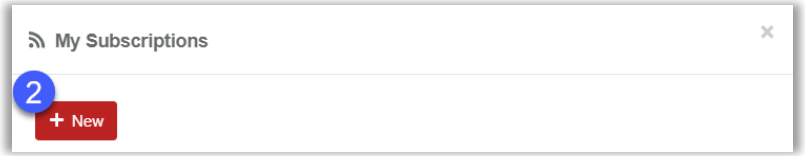
1. Log in to PCDC



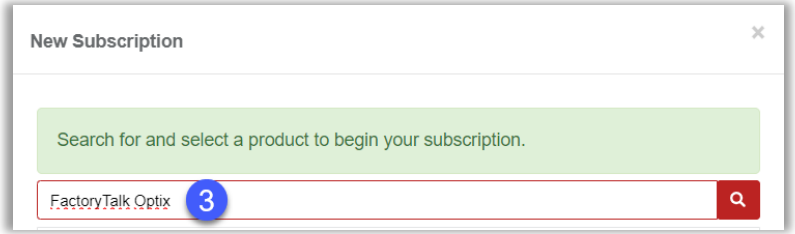
and on the right pane
select “Subscriptions”



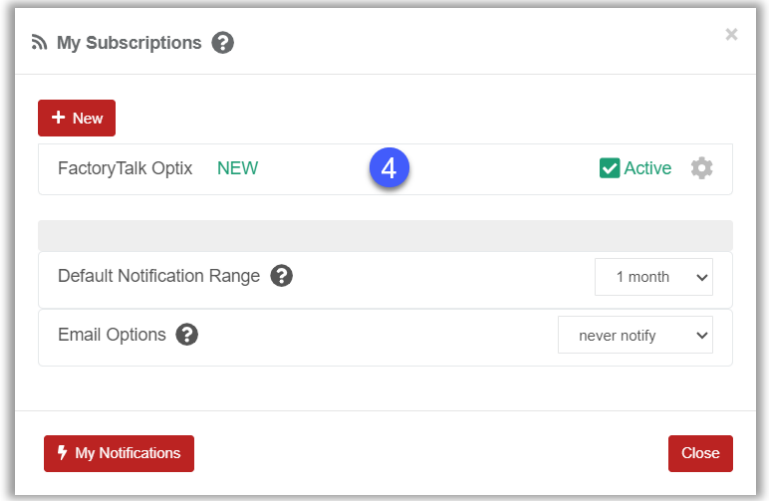
2. Click on **+ New** button



3. Search for "FactoryTalk Optix" and select it



4. FactoryTalk Optix
is added to your subscription list



5. With the settings button ⚙ you can select
the notification types that interest you,
and with the Email Options you can be notified via email
when you have new notifications

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Thank you

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